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# The Impact of the Future Pensions Act on the Purchasing Power of Dutch Pensions: A Numerical Approach

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**Abstract**

We address the life cycle investing problem in the context of the Dutch Future Pensions Act (FPA). Specifically, we examine the welfare loss and preservation of purchasing power in a defined contributions (DC) setting with a random process for realized and expected inflation. We use numerical optimization to find the optimal investment and consumption decisions for an investor and pension fund in a model with and without inflation. A key finding of our research is that the welfare loss due to FPA regulations can be reduced by approximately 1.4 percentage points when the law on determining the projected return would be relaxed. Incorporating expected inflation in the decision on pension disbursements would substantially benefit the purchasing power of the participant during retirement. Moreover, we find that the relaxation of the borrowing constraint to 150% nearly halves the corresponding welfare loss, but this is negligible compared to the sub-optimal projected return in the DC setting.

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# Chapter 1

# Introduction

On July 1, 2023, the Future Pensions Act (FPA) (Dutch: Wet Toekomst Pensioenen, WTP) came into effect in the Netherlands. This new law allows pension funds to use the returns from their investments more quickly to increase pensions. Moreover, it ensures that it becomes more transparent and personal how much pension has been accumulated per individual. Currently, the Netherlands is still in a transition phase, with a deadline on January 1, 2028. Employers, labor unions, and pension funds are discussing how they can modify their regulations to comply with the new law.

In this paper we extend the classic portfolio and consumption optimization problem by Merton (1971), by adding stochastic processes for expected and realized inflation. In the setting of the Dutch market and the regulations of the FPA, we examine how allowing to take expected inflation into account affects decisions on consumption. We compare the resulting overall welfare of different FPA regulations to the optimal situation without any restrictions.

With our research we aim to address the recurring concern on the ability of the pensions to preserve purchasing power under the FPA. On June 20, 2024, there was a meeting on this topic in the Dutch House of Representatives (Tweede Kamer der Staten-Generaal, 2024). During the meeting multiple position papers submitted by various pension fund advisors were discussed. Most of them conclude that despite the alleviation of the borrowing constraint to 150%, there is no guarantee that the FPA protects against inflation in all scenarios. Recently, the Dutch Minister of Social Affairs and Employment stated in Ministerie van Sociale Zaken en Werkgelegenheid (2025a) that there are still ongoing researches on the preservation of purchasing power under the new pension law. We focus on variant 2a, which was initially suggested by De Groot (2024), to increase flexibility by allowing the incorporation of expected inflation derived from the market when paying out pensions. This variant has several advantages: it avoids intergenerational redistribution, can be implemented after the transition without affecting it, has no impact on the government budget, employers, or pension contributions, and, when combined with variant 1, improves alignment with realized inflation, enhancing transparency and participant support.

Our approach diverges from the existing life cycle investing literature in several ways. First, despite having an additional state variable for the expected rate of inflation, we do not use a simulation-based approach like Koijen et al. (2010). Rather, we keep to the technique numerical dynamic programming using Gaussian quadrature like Cocco et al. (2005). Second, to adhere to the FPA, we use a borrowing constraint that still partly allows borrowing. Most literature that

includes a borrowing constraint does not allow borrowing by using a constraint of 100%. Third, we aim to narrow the gap between the theoretical model and its practical application to pension funds by adopting an alternative interpretation of the decision variables. In contrast to much of the existing literature, we do not frame the problem from the perspective of an individual investor, but rather focus on capital accumulation before retirement and pension disbursements thereafter. This corresponds to a defined contributions (DC) setting.

We consider the standard life cycle investing problem and extend it by incorporating expected and realized inflation in a DC setting. We optimize the model numerically using techniques as backward induction, endogenous state-space grids and Gaussian quadrature. We compare the numerical optimization of our standard model to the analytical solution from Merton (1971) and see similar results. For both models we compute the welfare loss resulting from a borrowing constraint of 150% and a projected return equal to the risk-free rate, both regulations in the FPA. For the model with inflation, we consider different values for the correlation between expected inflation and stock returns.

A key insight from this paper, resulting from Monte Carlo simulations in our optimal model with inflation, is that relaxing the law on determining the projected return can circumvent a welfare loss of a substantial 1.4%, when keeping the borrowing constraint at 150%. This results from the fact that the FPA projected return does not take time preference, returns on risky investments and expected inflation into account. A more positive value for the correlation reduces this potential welfare gain, but it remains of the same order of magnitude. Avoiding the welfare loss can be partly realized by a fixed mark-up or markdown on the projected return by the pension fund in the current legislation. However, large pension funds like ABP choose not to use this (ABP, Raad voor het Overheidspersoneelsbeleid, 2024), and to fully circumvent this welfare loss, the FPA legislation should be revised to permit a time-varying markup or markdown on the projected return based on expected inflation. Another finding is that the welfare that is lost due to the borrowing constraint of 150% is only approximately 0.03% in a DC setting. However, keeping the borrowing constraint at 100% nearly doubles this welfare loss.

We conclude that allowing pension funds to take expected inflation into account when disbursing to their participants would substantially improve the welfare of the Dutch citizen. This policy is better able to preserve the purchasing power of the participant during their retirement.

The remainder of this paper is organized as follows. Chapter 2 elaborates on the FPA and the existing literature for life cycle investing. We present our models and numerical optimization methods in Chapter 3. In Chapter 4, we report the parameters for these models that we use and some corresponding processes. Chapter 5 provides the results for our Monte Carlo simulations and resulting welfare losses. Chapter 6 concludes. Finally, Appendix A and B contain a proof and some additional figures.

# Chapter 2

# Literature Review

## 2.1 Purchasing power in the Dutch pension reform

During the meeting on June 20, 2024, multiple position papers on the purchasing power of Dutch pensions submitted by various pension fund advisors were discussed. Most of them conclude that despite the alleviation of the borrowing constraint to 150%, there is no guarantee that the FPA protects against inflation in all scenarios. For example, Mehlkopf (2024) outlines a scenario of stagflation: the purchasing power of pensions decreases and cannot be preserved by the underwhelming returns on risky assets. The same happens in a scenario of constant economic growth and high inflation: satisfactory stock returns cause the pensions to increase, but are insufficient to keep up with the inflation. The recent concerns on the FPA are therefore not surprising, given that we now find ourselves in a similar situation.

One of the solutions that came out the position papers is the utilization of the solidarity reserve (Dutch: solidariteitsreserve). This is a new instrument in the FPA, which acts as a buffer with various advantageous deployment options (Mehlkopf et al., 2021). In the recent experience of Mehlkopf (2024), he observes that the majority of pension funds determine to only use this reserve to prevent pension cuts, and not in the case of inflation risk. The FPA encourages this, by only allowing to use the reserve in case of unexpected (high) inflation. De Groot (2024) suggests to also legally permit pension funds to utilize the reserve to hedge against expected inflation in case of disappointing excess return. A downside of this is that the reserve is limited and eventually runs out when the period of high inflation persists. Moreover, the reserve is less powerful for funds with relatively more old participants.

Another solution is investing in index-linked bonds. According to Campbell & Viceira (2001), access to inflation-indexed bonds can generate a certainty-equivalent wealth effect of up to 10–30% for a risk-averse long-term investor. However, as stated by Mehlkopf (2024) and Bovenberg & Mehlkopf (2014), the markets for inflation-linked assets are typically not liquid, especially for large pension funds. Moreover, such markets that are more liquid are linked to inflation in for example the United States or United Kingdom, which may differ from the index that is relevant for a Dutch pension fund.

Mehlkopf (2024) suggests adding an instrument to the FPA that allows redistributing the bad consequences of inflation between age cohorts. This is based on the idea that old participants experience more harmful consequences from inflation than the young, since they are less able

to hedge this with future income or investment returns. However, the implementation is quite complex, and during long-lasting periods of inflation the young participants can bear even more dreadful effects.

The solution that we further investigate in this paper is the last one by De Groot (2024), in which the FPA allows the use of a real projected return (Dutch: projectierendement). The Dutch Minister of Social Affairs and Employment recently reported this solution in Ministerie van Sociale Zaken en Werkgelegenheid (2025a) (variant 2a) as a possible instrument that should be further investigated. The projected return determines how quickly a pension is paid out to a retiree. When higher returns are expected, the pension fund is able to distribute more, since the paid out assets are expected to be earned back in the future. The FPA obliges pension funds to take a projected return based on the nominal risk-free rate (Ministerie van Sociale Zaken en Werkgelegenheid, 2025b). For example, the large Dutch pension fund ABP states in its transition plan that it uses a projected return equal to the RTS published by De Nederlandsche Bank (DNB), without applying any upward or downward adjustment (ABP, Raad voor het Overheidspersoneelsbeleid, 2024). The use of the RTS in the projected return is a sub-optimal strategy, because it does not take rate of time preference, returns on investments and expected inflation into account. Hence, De Groot (2024) proposes to permit pension funds to incorporate market-derived expected inflation during the establishment of the projected return, as this will protect against inflation during the payout phase. Additional advantages of this instrument are that it avoids intergenerational redistribution, can be implemented after the transition without affecting it, has no impact on the government budget, employers, or pension contributions, and, when combined with variant 1, improves alignment with realized inflation, enhancing transparency and participant support.

## 2.2 Life cycle investing

The problem described above could be traced back to the classic long-term optimal investment and consumption decision problem, also known as life cycle investing. The problem was first introduced in a continuous-time setting by Merton (1969) and later, by using a Geometric Brownian motion for the risky assets, solved analytically by himself (Merton, 1971). He finds that the maximization of the constant relative risk aversion utility function, without any borrowing constraints, leads to an optimal fraction of wealth invested in risky assets that is independent of the level of wealth and age of the investor. Various researchers have extended the literature for this model by incorporating instruments to better represent the real world.

Bodie et al. (1992) analyze how individuals adjust their labor supply in response to changes in investment opportunities and wage uncertainty. The study finds that greater labor flexibility allows for more aggressive portfolio choices, as human capital acts as a buffer against financial risks. Since the fraction of human capital over financial capital is larger for the young than for the old, this suggests that young individuals should invest a higher share of their capital in risky assets.

Regarding labor income risk, Cocco et al. (2005) find that by adding a small probability of a substantial decrease in income to the life cycle model reduces the average share of capital invested in risky assets. However, a stochastic labor income that is uncorrelated with equity

returns increases the allocation to equities, especially at a younger age.

Cocco & Gomes (2012) evaluate the impact of longevity risk on the optimal life cycle investing and saving decisions. They find that an individual tends to save more during their lifetime for their retirement and decides to retire at an older age. Even after hedging longevity risk using an optimal saving and retiring strategy, the option to invest in longevity risk insuring financial assets would still be beneficial.

In a long-term investing setting such as the life cycle investing problem, interest rate risk should also be taken into account. As such, both Detemple & Rindisbacher (2005) and Liu (2007) find that stochastic interest rates significantly affect optimal portfolio decisions.

## 2.3 Inflation risk

Brennan & Xia (2002) introduce correlated Ornstein-Uhlenbeck processes for the interest rate and expected rate of inflation. They obtain simple closed-form solutions for the optimal investment strategy and indirect utility. In the presence of short-selling constraints, the optimal portfolio includes cash, equity, and a single nominal bond with a maturity that is optimally chosen based on the investor's investment horizon and degree of risk aversion. Campbell & Viceira (2001) use a similar model for inflation and conclude that long-term bonds can be attractive for infinitely-lived investors, even if they carry inflation risk in the short term.

Building on these two papers, Koijen et al. (2010) extend the models by using a similar factor structure but allowing for time-varying bond risk premia. This last model is commonly used as the standard model for Asset Liability Management analysis in the Dutch pension fund industry (Bouwman & Lord, 2016).

When incorporating an inflation process in the life cycle investing problem, the correlation between innovations in stock and bond returns and in inflation should be taken into account. Both Brennan & Xia (2002) and Koijen et al. (2010) find a negative correlation between the two. Campbell & Viceira (2005) report a negative correlation on the short term, but a positive correlation on the long horizon. Antonakakis et al. (2017) find, however, that for some years the correlation is significantly positive for the United States.

Since we aim to consider the model in the setting of the FPA, we take short-selling and borrowing constraints into account. Following from Merton (1971) and Bodie et al. (1992) and visualized by Bovenberg & Mehlkopf (2014), younger generations should optimally have an exposure of over 100% in risky assets. In the FPA, the borrowing constraint is alleviated from 100% to 150%, possibly resulting in pension funds choosing for the maximal level allowed given the constraint for their younger participants (Blake et al., 2014). This means the young have to borrow from the old, since pension funds are not allowed to be net borrowers.

By incorporating inflation risk in a life cycle investing problem with an alleviated borrowing constraint, we attempt to fill the gap in the literature, since this is not researched yet, to our knowledge. Moreover, we evaluate multiple scenarios by considering various signs and magnitudes for the correlation between expected inflation and stock returns. We solve all our models numerically, since Brennan & Xia (2002) state that it can not be solved analytically when borrowing constraints are involved.

# Chapter 3

# Methodology

In this chapter we discuss which methods and techniques are applied in our research. Section 3.1 explains the theory of life cycle investing and the considered model extensions. The second section describes the methods that we use to numerically optimize the life cycle models.

## 3.1 Life cycle investing

### 3.1.1 Standard Merton life cycle investing problem

We use the standard Merton (1971) life cycle model and its analytical solution as a benchmark model. This model maximizes the expected discounted utility of an individual during their working life in a continuous time setting. Without any bequest motives, following the notation from Bovenberg & Mehlkopf (2014), the investor solves

$$\max \mathbb{E}_t \left( \int_t^D e^{-\delta(s-t)} u(C_s) ds \right), \tag{3.1}$$

where $\mathbb{E}_t$ denotes the expected value at time $t$ and $\delta > 0$ represents the rate of time preference. The investor passes away at time $D$, which can be random. In our model, the first time period $t = 0$ represents the start of the investor's working career. The consumption at time $t$ is given by $C_t > 0$ and the utility function for consumption is given by

$$u(c) = \frac{c^{1-\theta}}{1-\theta}, \tag{3.2}$$

where $\theta > 1$ denotes the relative risk aversion coefficient. This utility function features positive and constant relative risk aversion (CRRA). At each point in time, the individual has to make a decision on what proportion of wealth is consumed and how much is invested in a risky asset and in a risk-free asset. Risk-free bonds $P_t$ and risky stocks $S_t$ develop over time as

$$\frac{dP_t}{P_t} = R_f dt, \tag{3.3}$$

$$\frac{dS_t}{S_t} = \mu_S dt + \sigma_S dZ_{S,t}, \tag{3.4}$$

where $R_f$ denotes the risk-free rate, $\mu_S > R_f > 0$ represents the expected returns on stocks, and $\sigma_S > 0$ is the standard deviation of the log stock returns. Most literature use the stock risk premium instead of the expected return on stocks, which is computed by $\lambda_S = (\mu_S - R_f)/\sigma_S$. $Z_{S,t}$ denotes the standard Brownian motion for the stock returns. The individual is also able to borrow money against the same interest rate that corresponds to the risk-free asset. The development of financial wealth $F_t$ is then given by

$$\begin{aligned} dF_t &= (1 - w_{S,t})F_t R_f dt + w_{S,t}F_t((R_f + \lambda_S\sigma_S)dt + \sigma_S dZ_{S,t}) + (Y_t - C_t)dt \\ &= (F_t - X_t)R_f dt + X_t\frac{dS_t}{S_t} + (Y_t - C_t)dt, \end{aligned} \tag{3.5}$$

where $w_{S,t}$ denotes the share of financial wealth invested in risky assets at time $t$ and $X_t \equiv w_{S,t}F_t$ represents the total financial assets in the stock portfolio. The income $Y_t$ is normalized such that $Y_t = 1$ for $t \in [0, T]$ and $Y_t = 0$ for $t \in (T, D]$, with $T$ being the retirement age. In the context of this benchmark model, human capital is riskless and equal to the present value of future income. Following Bovenberg & Mehlkopf (2014), we denote human capital by

$$H_t = \frac{1}{R_f}(1 - e^{-R_f(T-t)}), \tag{3.6}$$

for time $t \in [0, T]$ and $H_t = 0$ for $t \in (T, D]$.

With these assumptions, a closed-form analytical solution exists, originally found by Merton (1969). Given that the total wealth $W_t$ is equal to the sum of financial wealth $F_t$ and human capital $H_t$, the optimal share of total wealth invested in risky assets is given by

$$\hat{w}_{S,t} = \frac{\lambda_S}{\sigma_S\theta}, \tag{3.7}$$

hence not being dependent on time and total wealth. This is also called the Merton fraction. Following Bovenberg & Mehlkopf (2014), the optimal consumption policy is given by

$$C_t = (F_t + H_t)g(D - t)^{-1}, \tag{3.8}$$

where the function $g(D - t)$ is solved for

$$g(D - t) = \frac{1}{A}(e^{A(D-t)} - 1), \tag{3.9}$$

with

$$A = -R_f + \frac{R_f - \delta}{\theta} + \frac{1}{2}\frac{1 - \theta}{\theta^2}\lambda^2. \tag{3.10}$$

The expression for $A$ can be interpreted as the negative of the projected return that is mentioned in Section 2.1. When higher returns are expected, an investor is able to consume more in the present, since the consumed assets are expected to be earned back in the future. We see this relation back in equations 3.8 and 3.9. However, as already mentioned in Section 2.1, the FPA obliges pension funds to take a projected return based on the nominal risk-free rate (Ministerie van Sociale Zaken en Werkgelegenheid, 2025b). This causes the consumption decision rule to be suboptimal, since the expression in equation 3.10 then becomes $A = -R_f$, which does not take

the time preference and risk premium parts into account.

### 3.1.2 Inflation risk

We extend the model by including a stochastic inflation rate. We follow Brennan & Xia (2002), by defining two separate stochastic processes for the realized rate of inflation and the instantaneous expected rate of inflation. The process for the price level $\Pi_t$ is given by

$$\frac{d\Pi_t}{\Pi_t} = \pi_t dt + \sigma_\Pi dZ_{\Pi,t}, \tag{3.11}$$

where $\sigma_\Pi > 0$ denotes the volatility for the price level and $Z_{\Pi,t}$ gives the standard Brownian motion for the price level evolution. The drift $\pi_t$ represents the expected rate of inflation, which follows an Ornstein-Uhlenbeck process that is given by

$$d\pi_t = \kappa_\pi(\overline{\pi} - \pi_t)dt + \sigma_\pi dZ_{\pi,t}, \tag{3.12}$$

where $\kappa_\pi > 0$ stands for the mean reversion speed of the expected inflation rate, $\overline{\pi} > 0$ represents the long-term expected rate of inflation, $\sigma_\pi > 0$ is the corresponding volatility and $Z_{\pi,t}$ is the standard Brownian motion for this process. $Z_{S,t}$, $Z_{\Pi,t}$ and $Z_{\pi,t}$ can be correlated, which is denoted by $\rho_{S,\Pi}$, $\rho_{S,\pi}$ and $\rho_{\Pi,\pi}$. The effect of the price level on the financial wealth of the investor can be computed by dividing by the uncertain rate of inflation as

$$dF_t = \left((F_t - X_t)r_t dt + X_t\frac{dS_t}{S_t} + (Y_t - C_t)dt\right) / \left(1 + \frac{d\Pi_t}{\Pi_t}\right), \tag{3.13}$$

which is not derived from Brennan & Xia (2002). This is due to the fact that they use pricing kernels and stochastic discount factors, which are not necessary for the numerical solution. Note that this implies that the realized inflation only has effect for one period and that for example the income $Y_t$ has increased with the inflation after that period. We aim to see what effect the knowledge on the expected inflation has on the optimal decisions for the investor over time.

### 3.1.3 Borrowing constraint

As described at the end of Section 2.3, the borrowing constraint for pension participants is alleviated from 100% to 150% in the FPA. All methods described above, assume there are no borrowing or short-selling constraints. Brennan & Xia (2002) state that when constraints are present, no analytical solution exists, implying that the optimal portfolio and consumption decisions have to be found numerically. For our model, this indicates that our maximization problem in equation 3.1, with financial wealth dynamics for $F_t$, is subject to the constraint

$$0 \leq X_t + C_t \leq \overline{w}(F_t + Y_t) \tag{3.14}$$

where $\overline{w}$ denotes the maximum share invested in risky assets. So for example, for the old pension system we have $\overline{w} = 1$, while for the FPA we have $\overline{w} = 1.5$. On top of the assets in the stock portfolio, we add the consumption amount in the constraint, since we assume the assets to consume also have to be borrowed. Additionally, we already add the income from year $t$ to the

financial wealth, because we do not consider it borrowing when the investor knows he or she collects that income certainly early in that year.

### 3.1.4 Defined contributions

The FPA is known as a defined contributions (DC) retirement plan. This means that the amount that is contributed by the investor to her pension is fixed. However, the amount that is received after retirement is not guaranteed, as this depends on the outcome of the investments of the pension fund. The old pension law was an example of a defined benefit (DB) retirement plan, where the payout after retirement is fixed beforehand. In DC, the investor carries the investment risks, not the employer or pension fund as in DB.

We aim to narrow the gap between the theoretical model and the DC retirement plan by adopting an alternative interpretation of the variables. Moreover, to implement the FPA projected return we need a formula for human capital, which we do not have in the model with inflation. We start by interpreting the income $Y_t$ as the annual labor income, from which a fixed amount is consumed and the remainder is added to the pension pot. We keep the annual labor income fixed at 1, so essentially the variable $Y_t$ does not change. The investment decision does not change as well, except for that this is now interpreted as the amount of the pension of the investor that is invested in stocks by the pension fund.

Lastly, the consumption decision $C_t$ is interpreted as the amount that is kept or paid out to the investor at time $t$. Before retirement, this is the same fixed amount every year, so we have $C_t = a$ for $t \in [0, T]$. The level of $a$ is determined by maximizing the utility after taking the optimal strategy for after retirement into account. This level is approximately 0.75 for all correlation and regulation combinations, meaning that 25% of the income is annually contributed to the pension of the investor. This savings rate can be compared to the contribution rate applied to the pension base (Dutch: pensioengrondslag) in real-world systems, where participants typically save a percentage of their income above the AOW threshold. The AOW is the basic state pension in the Netherlands, and income below this threshold is excluded from pension accrual. Since our model excludes AOW and applies the savings rate to total income, the 25% is roughly comparable. This savings rate is roughly in line with the contribution rates used by major Dutch pension funds. For example, the education sector pension fund ABP applies a total contribution rate of 27% of the pension base (PO-Raad, 2024), while the healthcare and welfare fund PFZW charges a total rate of 25.8% (PFZW, 2024). This fixed percentage of the income contributed to the pension changes the problem, as there is no variable consumption before retirement. We only implement this DC interpretation for the model with inflation.

## 3.2 Numerical optimization

This section elaborates on the methods we use for the numerical optimization. Our approach is most similar to the one of Cocco et al. (2005). Each subsection dives deeper into each method. For our numerical optimization we discretize to periods of one year, making $dt = 1$.

### 3.2.1 Backward induction

Like most papers on numerically optimizing life cycle investing decisions, we use backward induction. It is clearly step-by-step explained in the appendix of Koijen et al. (2010). Backward induction breaks the problem in discrete time periods and optimizes for each time period from end to start. Assuming the investor has no bequest motives, we know that for the last period $D$ the investor consumes all available wealth, hence $C_D = W_D = F_D + H_D = F_D$ (since $H_D$=0). In this last period the portfolio decision $w_{S,D}$ is irrelevant. We calculate the utility of the investor corresponding to the level of wealth using $V_D(F_D, \pi_D) = u(C_D) = u(F_D)$. Then, for each period before we solve the Bellman equation (see e.g. Carroll (2006), Koijen et al. (2010))

$$\max_{w_{S,t}, w_{P,t}, C_t} V_t(F_t, \pi_t) = \max_{w_{S,t}, w_{P,t}, C_t} u(C_t) + \left( \sum_{i=1}^{D-t} e^{-\delta i} \right) \mathbb{E}(V_{t+1}(F_{t+1}, \pi_{t+1})), \tag{3.15}$$

where $F_{t+1}$ and $\pi_{t+1}$ are stochastic and can be determined using the processes in equation 3.5 or 3.13 and equation 3.12, respectively. These two variables financial wealth and expected inflation are together with time the three state variables in this problem. For the standard Merton model without inflation the expected inflation is not included as a state variable. The summation in equation 3.15 represents the addition of all future discount factors, which can not be multiplied since we are using discrete time periods. After finding the optimal decision variables, we save the associated maximum value for each time period together with the consumption equivalent given by

$$CE_t = u^{-1} \left( V_t(F_t, \pi_t) / \left( \sum_{i=0}^{D-t} e^{-\delta i} \right) \right), \tag{3.16}$$

where $u^{-1}(\cdot)$ is the inverse of the utility function in equation 3.2. The calculation and saving of this variable is crucial for the next step.

### 3.2.2 State-space grid and interpolation

At each period in time, there is a set of variables that have to be given since they can not be determined based on other variables. This is the case for the endogenous variable $F_t$, since the wealth is dependent on decisions taken in the past. Next to that, at time $t$ it is unclear how the macro variable expected inflation $\pi_t$ has developed over time. For these variables we construct endogenous state grids, following the methods of Carroll (2006). For each point in time, we construct a grid of $K$ levels of wealth and $L$ levels of expected inflation. These grids are based on prior knowledge about the dynamics of each state variable, making sure that the eventually simulated paths will not exceed the boundaries of the grid. Next to that, the grids are more dense at more interesting states. The decision variables are optimized for each combination of state variables, resulting in $D \times K \times L$ optimization iterations in total. This grid is used in two ways.

The first use-case of the grid is during the optimization of equation 3.15. After computing the expected value for $F_{t+1}$ and $\pi_{t+1}$, it is likely that these values or not exactly present on the grid. That is why we interpolate the relation between the state variables and the consumption equivalent for period $t+1$, which we saved for each gridpoint. Using the expected values for the

state variables for the next period, we can compute the expected consumption equivalent. From this we can use the utility function to find the expected value, which can then be filled in for the Bellman equation. Instead of interpolating the relation for the value function directly, we use the consumption equivalent because this relation is more smooth and better interpretable.

For the standard model we use a linear interpolation method that allows for extrapolation, since we know that in this model the relation between future assets and future consumption equivalent is perfectly linear. For the model that includes inflation, we also use linear interpolation that allows for extrapolation, but using a two-dimensional input. The relation between future assets, future expected inflation and future consumption equivalent is not linear (see Figure B.1 in Appendix B), but this does not cause problems due to the density of our grid. Moreover, an interpolator that allows for non-linear relations extremely increases computational time.

After maximizing the utility for the investor for each gridpoint, we simulate $N$ market scenarios using the dynamic processes described in Section 3.1. Each simulated path is evaluated for all time periods by the interpolated relations between the state variables and the decision variables. By taking the median and some percentiles, we are able to see the dynamics of the optimal decision variables over time.

### 3.2.3 Gaussian quadrature

To calculate the expected value in equation 3.15, an integral has to be computed for each random variable that the future assets depend on. Doing this for each optimization iteration asks for immense amounts of computational time. That is why, similar to Cocco et al. (2005), we approximate the integrals using Gauss-Hermite quadrature methods. For our random variables we have

$$\mathbf{Z} = \begin{bmatrix} dZ_{S,t} \\ dZ_{\Pi,t} \\ dZ_{\pi,t} \end{bmatrix} \sim \mathcal{N}(0, \Sigma), \tag{3.17}$$

where $\Sigma \in \mathbb{R}^{3\times 3}$ is the covariance matrix. Let $G$ be the Cholesky decomposition of $\Sigma$, such that $\Sigma = GG^\top$ and

$$\mathbf{Z} = G \begin{bmatrix} \varepsilon_{S,t} \\ \varepsilon_{\Pi,t} \\ \varepsilon_{\pi,t} \end{bmatrix} \quad \text{with} \quad \varepsilon_{i,t} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0,1), \tag{3.18}$$

transforming our set of random variables into a linear combination of independent standard normals. For the approximation of the expected value we use $n$ Gauss-Hermite weights $w_i, w_j, w_k$ and corresponding nodes $\mathbf{x} = [x_i, x_j, x_k]^\top$. The approximation becomes

$$\mathbb{E}[f(\mathbf{Z})] \approx \frac{1}{\pi^{3/2}} \sum_{i=1}^{n} \sum_{j=1}^{n} \sum_{k=1}^{n} w_i w_j w_k f\left(\sqrt{2} G \mathbf{x}\right), \tag{3.19}$$

which is proven in Appendix A.1. In our case, $f(\mathbf{Z})$ is the consumption equivalent for future assets.

### 3.2.4 Welfare loss

As seen above, there are several regulations in the FPA that constrain pension funds from choosing the optimal consumption and investing decisions. For each regulation we aim to compute the total welfare that is lost by the implementation of it. The first regulation we consider is the borrowing constraint of 150%. The second regulation is taking the projected return equal to the nominal risk-free rate, without taking future returns on risky investments and inflation into account.

Following Bovenberg et al. (2007) and related studies, we measure welfare by calculating the certainty equivalent consumption level associated with the total utility over lifetime. We prefer this measure over raw utility values, as differences in utility are not directly interpretable due to the power form of the utility function. For each simulation path $i$, the total utility is computed by

$$U_i = \sum_{s=0}^{D} e^{-\delta s} u(C_{s,i}), \tag{3.20}$$

where $C_s$ adheres to the different suboptimal and optimal consumption rules. We take the average utility $\overline{U} = \frac{1}{N}\sum_{i=1}^{N} U_i$ and compute the constant consumption level that is equivalent to this level of average utility by

$$CE = u^{-1}\left(\frac{\overline{U}}{\sum_{t=0}^{D} e^{-\delta t}}\right) = \left(\frac{(1-\theta)\overline{U}}{\sum_{t=0}^{D} e^{-\delta t}}\right)^{\frac{1}{1-\theta}}, \tag{3.21}$$

where $u^{-1}(\cdot)$ is the inverse of the utility function in equation 3.2.

# Chapter 4

# Data

## 4.1 Parameter settings

In our model, we need to set multiple parameters. All parameters can be found in Table 4.1. They are based on values commonly found in the literature, and partially selected for computational convenience, to ensure numerical tractability and interpretability of results. The borrowing constraint $\overline{w}$ has multiple values, with $\overline{w} = 1$ corresponding to the old pension law, $\overline{w} = 1.5$ to the new FPA, and $\overline{w} = \infty$ to compare to a scenario where there is no borrowing constraint. We set the market parameters such that the long term real interest rate $(R_f - \overline{\pi})$ is equal to the rate of time preference $(\delta)$. As explained earlier, there is no clear value for the correlation between stock returns and expected inflation resulting from literature. Most literature finds a negative value, but there are some examples that show a positive correlation. For that reason we try three different values for this correlation, to ensure that our results hold in each of these situations. We assume that the unexpected component of the inflation has no correlation with the other variables, such that it is completely unpredictable.

Table 4.1: Parameter settings

| Parameter | Value | Parameter | Value | Parameter | Value |
|---|---|---|---|---|---|
| **Investor characteristics** | | **Stock market** | | **Inflation** | |
| $D$ | 60 | $R_f$ | 0.04 | $\sigma_\Pi$ | 0.01 |
| $T$ | 40 | $\lambda_S$ | 0.15 | $\kappa_\pi$ | 0.2 |
| $\delta$ | 0.02 | $\sigma_S$ | 0.2 | $\overline{\pi}$ | 0.02 |
| $\theta$ | 5 | | | $\sigma_\pi$ | 0.015 |
| **Correlations** | | **Numerical optimization** | | **Borrowing constraint** | |
| $\rho_{S,\Pi}$ | 0 | $K$ | 100 | $\overline{w}$ | $1, 1.5, \infty$ |
| $\rho_{S,\pi}$ | -0.3, 0, 0.3 | $L$ | 90 | | |
| $\rho_{\Pi,\pi}$ | 0 | $N$ | 50000 | **Discretization** | |
| | | $n$ | 12 | $dt$ | 1 |

## 4.2 Processes

Using Monte Carlo simulations, we can see the expected development of the dynamic processes over time. This can be useful for the process of expected inflation in determining the boundaries of the state grids. Using the parameter settings given above, we get an insight in the dynamics of the expected inflation by looking at Figure 4.1.

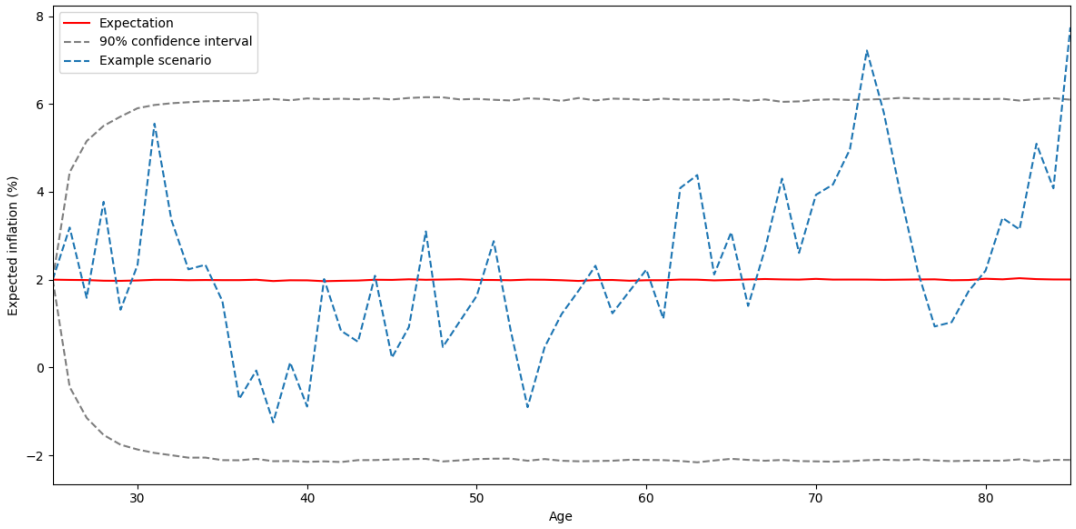



Figure 4.1: Expected inflation

We see that there is a 90% chance that the expected inflation does not exceed a value of approximately $-2\%$ and $+6\%$. This informs us to choose the boundaries of our state grid for $\pi_t$ slightly larger than this, such that our simulations will not reach values outside our grid.

The resulting price level has an extra unexpected component. We show the evolution of the price level over time in the figure below.

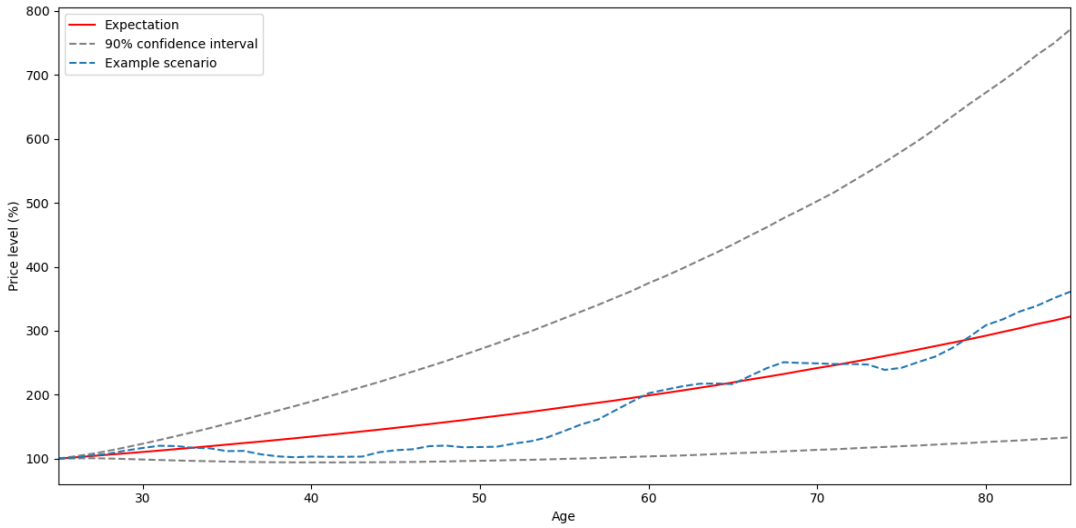



Figure 4.2: Price level

Figure 4.2 shows an increasing trend, which comes from the long-term expected inflation of 2%. In 60 years, the expected value of the price level is more than three times the original price

level.

We do not show the process for the stock market, because this would just be a random walk with a trend, which has no memory. The process for financial wealth depends on the decision variables and will be discussed in Chapter 5.

# Chapter 5

# Results

## 5.1 Life cycle investing without inflation

First, we consider the model without inflation. Financial wealth follows the process in equation 3.5. We compare our numerical optimized solution to the analytical solution from Merton (1971). Afterwards, we look at the dynamics of different variables and welfare loss under different regulations.

### 5.1.1 Analytical versus numerical

Our numerical optimal solution to the life cycle investing problem without inflation should show the same results as the analytical. We compare the numerical optimal consumption and stock decisions to the closed-form solutions reported by Bovenberg & Mehlkopf (2014). First, we discuss and compare the evolution of consumption for an investor over her lifetime to the dynamics of the analytical solution reported in equation 3.8. For both, we show the graphs for the expectation and the 5% and 95% quantiles together in Figure 5.1.

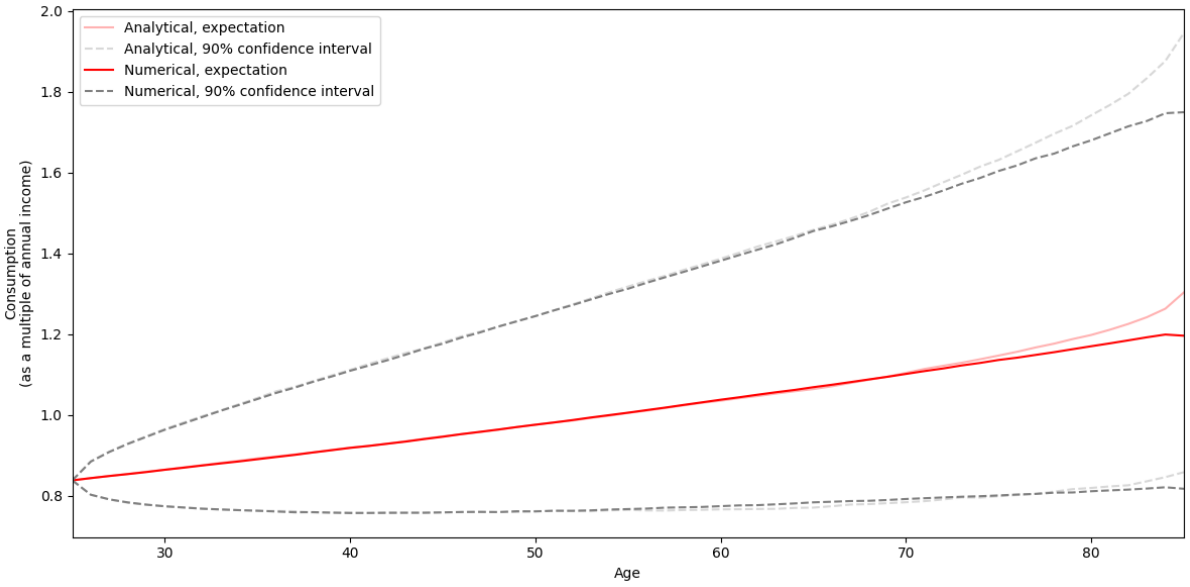



Figure 5.1: Consumption level in analytical and numerical solution

Figure 5.1 shows that for all scenarios, the initial consumption is equal, as there is not uncertainty at this point. Then, for the average investor the consumption gradually increases over her lifetime, almost linearly. This is as expected, since the goal of the investor is to smooth out the level of consumption during her life. Moreover, the fact that the rate of time preference is smaller than the interest rate, which causes the investor to prefer saving over consuming, explains the increasing trend. In the better scenarios, as seen by looking at the 95% percentile, we see that the level of consumption at a certain point even exceeds the level of labor income. This suggests that in events of high returns on the stock market, the investor is able to consume more by using her profit from her investment decisions.

When we compare the analytical solution with the numerical, we see a clear resemblance. This suggests that our optimal numerical solution is in line with the analytical solution in equation 3.8. The only observable difference is at the end of the lifetime of the investor, where the analytical solution shows an increasing slope. This follows from the fact that in the analytical model, the investor can continue to invest and consume continuously up to the final moment. In the numerical model, however, the investor only makes decisions at discrete time points (yearly), and no decisions at the final period $T$, since we treat it as the terminal state. Therefore, consumption and investment at the very end of the horizon are not modeled. Additionally, in the analytical solution the investor can possibly consume more than her amount of financial wealth in the last period, which we do not allow.

To compare our solution to the solution from Merton (1971), we also look at the Merton fraction. The Merton fraction, given in equation 3.7, shows that the share of total wealth invested in stocks does not depend on the level wealth nor time. We analyze this by looking at the graph for the share of total wealth invested in stocks, where the total wealth is the sum of financial wealth and human capital minus consumption. The result is shown in the Figure 5.2.

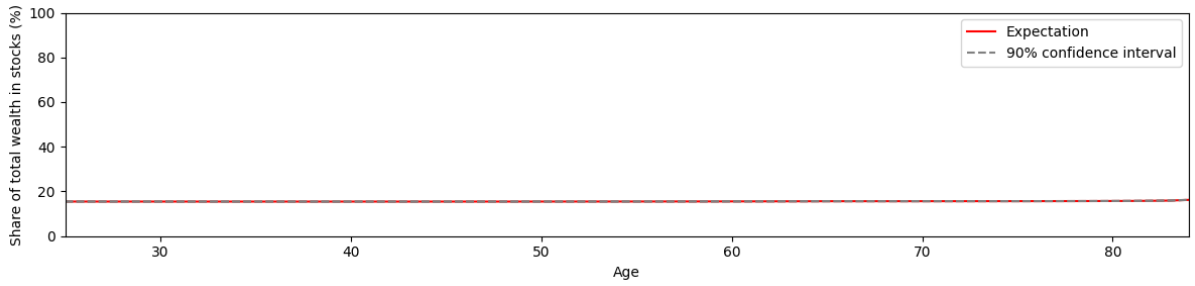


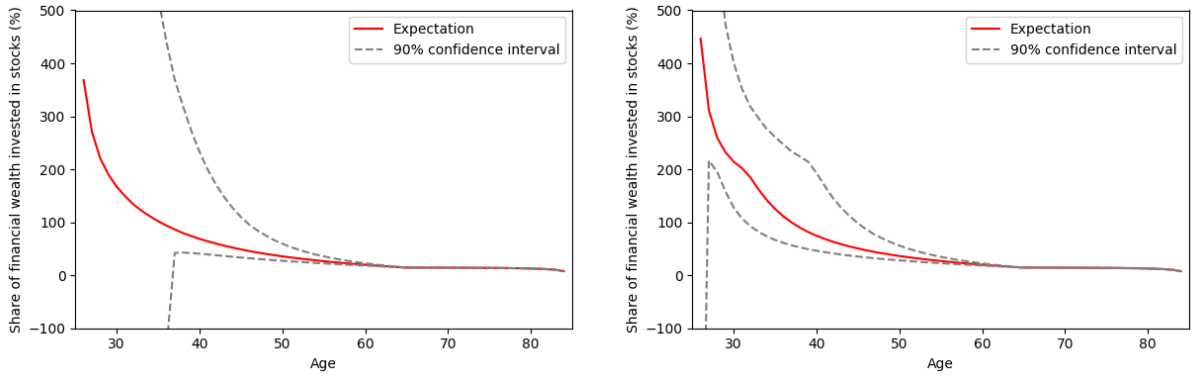

Figure 5.2: Share of total wealth invested in the stock portfolio

Figure 5.2 clearly shows a straight line over the complete lifetime of the investor, for all scenarios. This implies that in our numerical solution, this fraction does not depend on time nor wealth, as in the analytical solution. Moreover, the level of the fraction in Figure 5.2 is 15%, which corresponds to the value that results from equation 3.7 when we fill in our parameters.

### 5.1.2 Stock portfolio

Next, we look at how the stock portfolio decision develops over time for the investor. We consider the regulation of the FPA that affects this decision the most, namely the borrowing constraint. We compare the optimal situation without any restrictions to the situation with a borrowing

constraint equal to 150%. A constraint of 150% corresponds to $\overline{w} = 1.5$ in equation 3.14. Figure 5.3 below shows the resulting effects.




(a) Optimal, $\overline{w} = \infty$

(b) $\overline{w} = 1.5$

Figure 5.3: Share of financial wealth in the stock portfolio over time under different borrowing constraints

First, we look at the common patterns on Figures 5.3a and 5.3b. Both figures show that the average investor has a high share of financial wealth invested in stocks during her early working life, even exceeding 100%. This result is well-known since the findings of Merton (1971). The negative share of assets for the 5th percentile does not indicate that the investor is short selling in such a scenario, but comes from a negative financial wealth. Both figures show the exact same pattern after the retirement age of 65, for all scenarios. This corresponds to the Merton fraction when consumption would be subtracted from the financial wealth, since total wealth and financial wealth are equal in this period.

A difference is that Figure 5.3a shows a smoothly decreasing share for the expectation and 95% quantile, while Figure 5.3b has a kink for all quantiles around the 200% level. This is surprising, because the graph exceeds the level of the level of the borrwoing constraint, 150%. This is due to the fact that equation 3.14 does not necessarily imply that $X_t \leq \overline{w} F_t$. The locations where the graph exceeds 150%, are scenarios in which the investor uses borrowed money to invest and consume when she has a small amount of financial wealth. Moreover, in this graph we do not consider the income of time $t$ as acquired yet, while the borrowing constraint allows the investor to invest or consume this anticipated income. When future income can not be considered in the constraint, the investor would not be able to invest or consume in the first period, since she would not have any financial wealth. Another difference we see between the two figures, is that in the scenario with a borrowing constraint, the graph is negative for a much shorter period. We will discuss this in Section 5.1.4.

To see if the investor adheres to the borrowing constraint, we instead look at the graph in Figure 5.4. This shows the fraction of financial wealth plus income that is allocated to the stock portfolio or is consumed. According to equation 3.14, this fraction should be capped by the level of the borrowing constraint.

Figure 5.4 indeed clearly shows a cap at the 150% level, indicating that the investor adheres to the borrowing constraint for all scenarios. The borrowing constraint is initially binding for all quantiles, but becomes non-binding at different points in the life cycle. After this point and

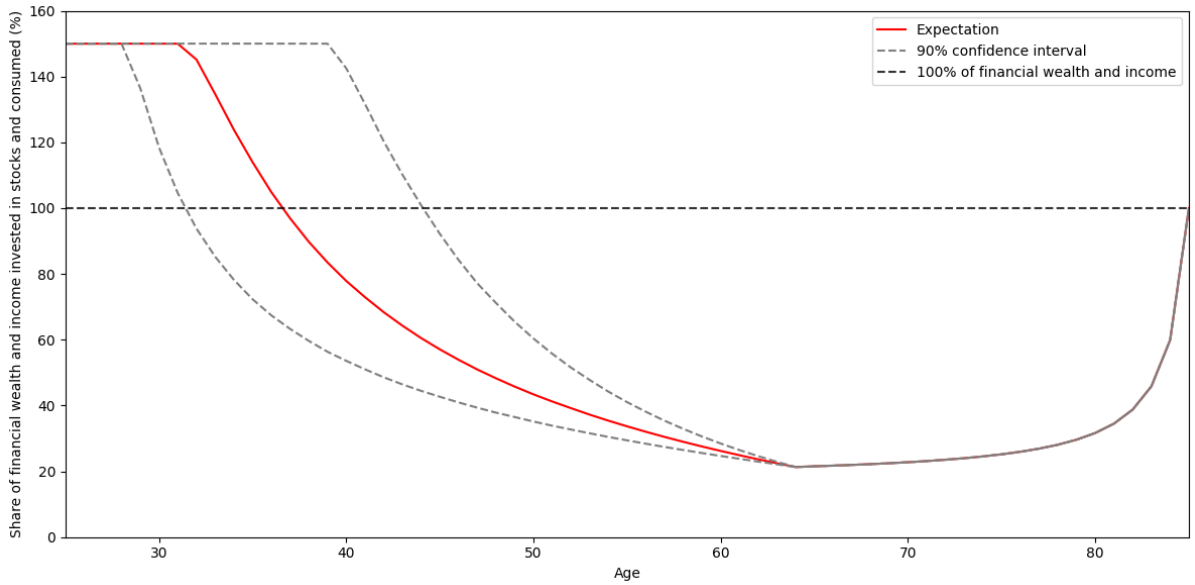



Figure 5.4: Share of financial wealth plus income allocated to the stock portfolio or consumed, with $\overline{w} = 1.5$

before retirement, the graph shows a similar pattern as in Figure 5.3. After retirement, the fraction is equal for all scenarios. It increases back to 100%, illustrating the consumption of the remaining financial wealth. When the share in Figure 5.4 is below 100%, the remainder of the financial wealth and income is invested in the risk-free bond.

### 5.1.3 Consumption

We have seen and discussed the evolution of the consumption decision in a scenario without restrictions in Section 5.1.1. Now, we consider a regulation of the FPA that affects this the most, namely taking the projected return equal to the risk-free rate. We optimize the stock portfolio, but consider the consumption as given, following equation 3.8 where $A = -R_f$. This does not include the time preference and risk premium part of equation 3.10, making it sub-optimal. We compare the non-restricted consumption evolution to the FPA consumption rule in Figure 5.5.

Figure 5.5 shows that the initial consumption level using the FPA projected return is higher than in the non-restricted scenario. If an investor expects higher future returns, she can consume more in the current period, so the graph suggests that the projected return for the FPA is higher than the optimal projected return. This is indeed the case, because filling in our parameters in equation 3.10 results in a projected return that is lower than the risk-free rate. However, as Figure 5.5 shows, this does not imply a higher consumption level for the remaining years. The projected return in the FPA is too high, causing the investor to excessively consume in the first years, resulting in a lower financial wealth later in life. Which of the two consumption graphs results in a higher welfare, we will discuss in Section 5.1.5.

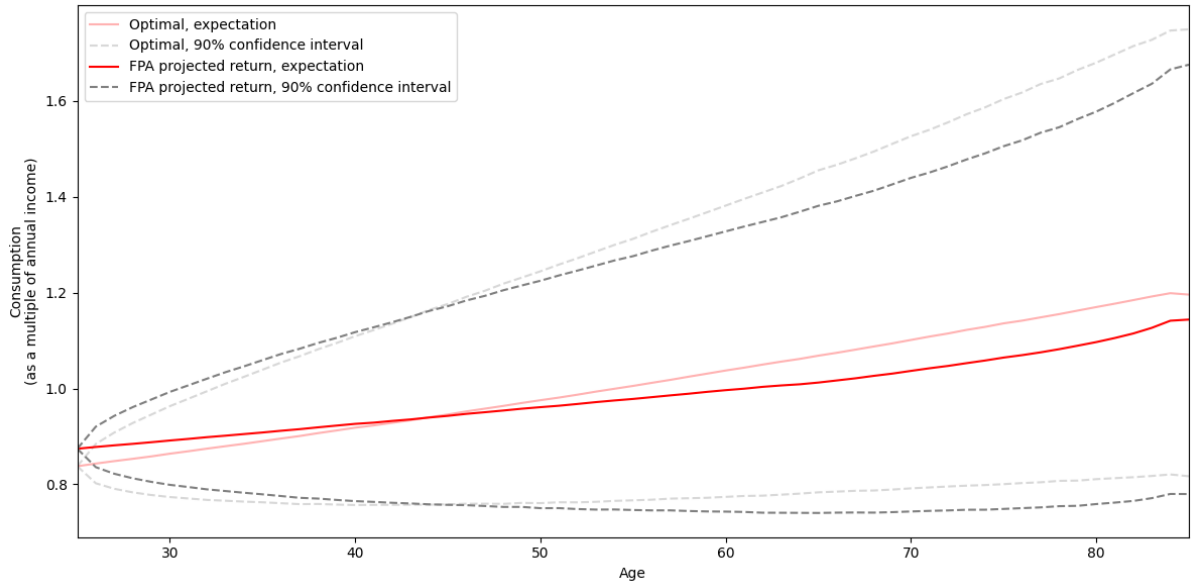



Figure 5.5: Consumption level in the optimal situation and with the FPA projected return

### 5.1.4 Financial wealth

The last variable we analyze is the evolution of the financial wealth of the investor. Both decision variables and all FPA regulations affect this variable. First, we look at the process of the financial wealth in the optimal situation, without any restrictions. The expectation, 90% confidence interval and two example scenarios for this variable are shown in Figure 5.6 below.

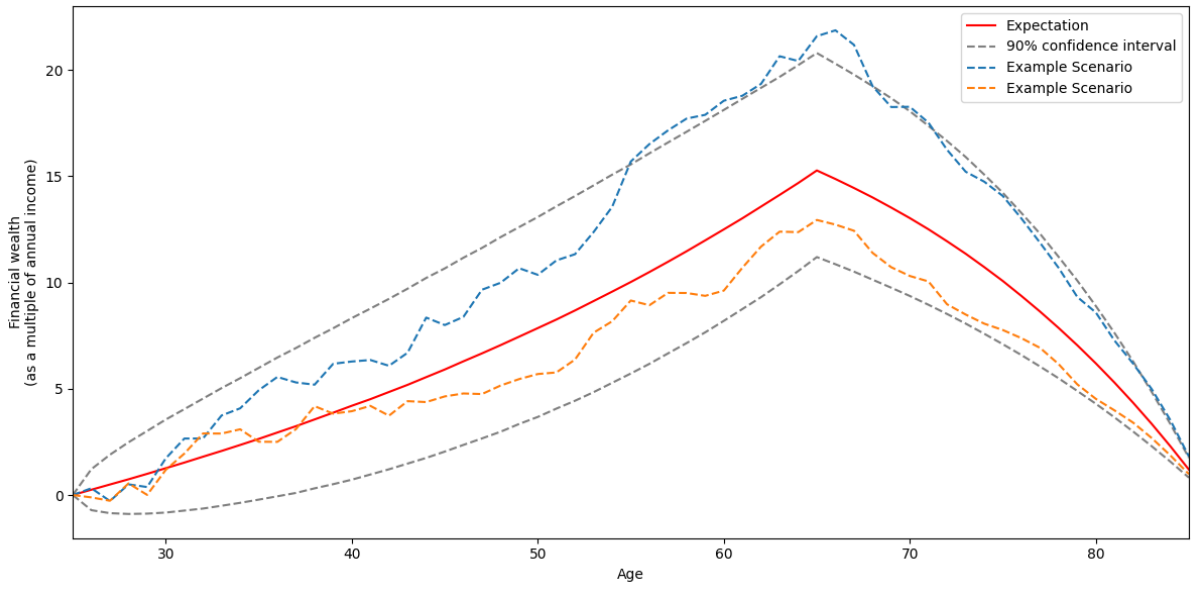



Figure 5.6: Financial wealth

Figure 5.6 shows that the financial wealth of the average investor increases during the period that she works and decreases from the point of retirement. In all scenarios, the wealth reduces back to 0 at the point of death, as is expected since we assumed the investor has no bequest utilities. As can be seen from the 5th percentile, there are certain scenarios for which the

financial wealth can become negative, meaning that the investor is in debt. This is caused by the lack of any constraints in this situation. The age at which the 5% quantile becomes positive, corresponds to the moment the 5% quantile in Figure 5.3 crosses the 0% level. Eventually, in all scenarios the investor has a positive amount of financial wealth at time of retirement. During retirement the level of wealth does not become negative. This can be interpreted as avoiding the risk of not being able to pay back any debt, as there is no certain labor income during this period. Lastly, the average amount of wealth shows a convex pattern before retirement. This is caused by the fact that we have chosen the parameters such that the interest rate is higher than the rate of time preference. This causes the investor to prefer saving over consuming, leading to a higher wealth and creating an increasingly steepening trend.

Next, we analyze how the two regulations of the FPA together affect the financial wealth. The borrowing constraint mostly affects the the stock portfolio decision as seen in Figure 5.3, while the projected return equal to the risk-free rate disrupts the consumption decision, as seen in Figure 5.5. We show the resulting evolution for the financial wealth in the FPA in Figure 5.7 below.

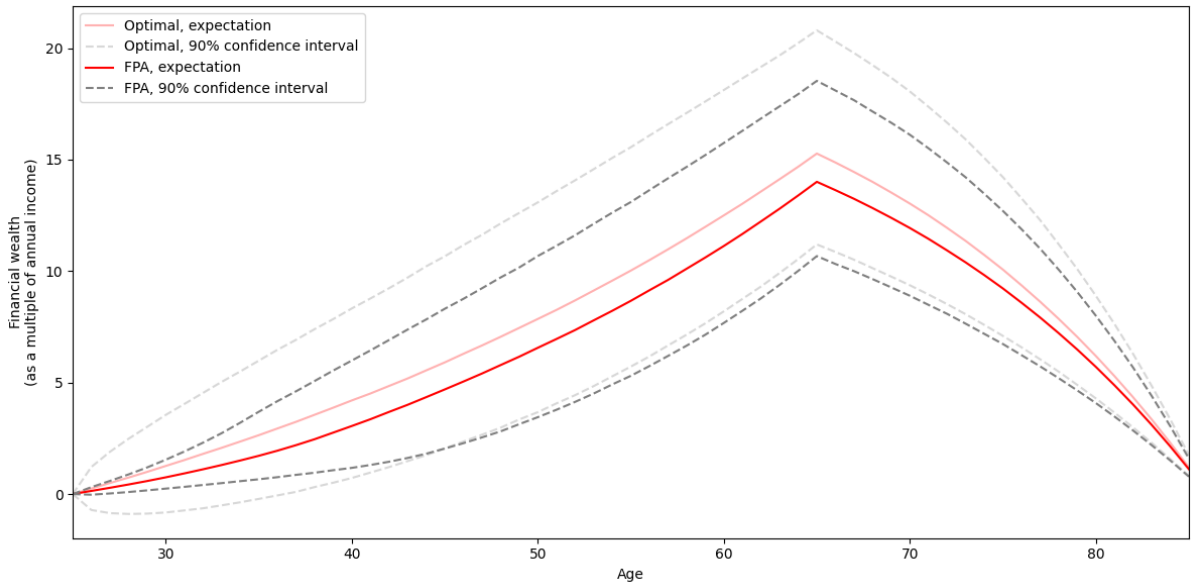



Figure 5.7: Financial wealth evolution in the FPA versus the optimal situation

The graph for the FPA regulations shows a similar pattern, with a convex increasing expectation before retirement and a concave decreasing evolution towards 0 after retirement. A difference between the two situations is that with the FPA, the confidence interval is more narrow. This is mostly caused by the borrowing constraint, which does not allow for too much wealth allocated to the stock portfolio, making the evolution of the financial wealth less risky. This is best observable in the early working life, due to the fact that during this period there is a greater chance to end up in debt, which is more difficult to pay back with a borrowing constraint. There are some simulations for which the financial wealth becomes negative, barely observable in Figure 5.7, but this is a much shorter period than in the situation without restrictions. Despite taking less risk in the early working life, for all three presented quantiles the investor ends up with a higher financial wealth at retirement when there are no restrictions.

This is partly caused by the borrowing constraint, but mostly by the sub-optimal consumption rule. As we have seen in Figure 5.5, the projected return of the FPA causes the investor to consume excessively in the early years, eventually ending up with less financial wealth later in life. In conclusion, the combination of the two regulations generates a more stable, but in most scenarios a lower outcome for the financial wealth of the investor.

### 5.1.5 Welfare loss

This section discusses how much welfare is lost as a result of sub-optimal laws in the FPA. Table 5.1 shows the results of the welfare losses under different scenarios. The first scenario is the standard optimal solution without any regulations, corresponding to Figure 5.6. The second and third scenarios have a borrowing constraint of 100% and 150%, respectively. In the fourth situation, the projected return is equal to the risk-free rate, as is proposed by the FPA. The last scenario combines the two laws of the FPA. Welfare is reported in the level of consumption that is equivalent to the average utility for investor, following equation 3.21. The loss is calculated as percentage with respect to the optimal scenario with no regulations.

Table 5.1: Welfare levels expressed as consumption equivalents under different regulations of the FPA without inflation

| | Optimal | 100% BC | | 150% BC | | RTS PR | | RTS PR & 150% BC | |
|---|---|---|---|---|---|---|---|---|---|
| CE | 0.919 | 0.911 | (-0.88%) | 0.915 | (-0.45%) | 0.916 | (-0.30%) | 0.909 | (-1.05%) |

**Note:** CE = consumption equivalent; BC = borrowing constraint; RTS PR = projected return equals RTS of the DNB.

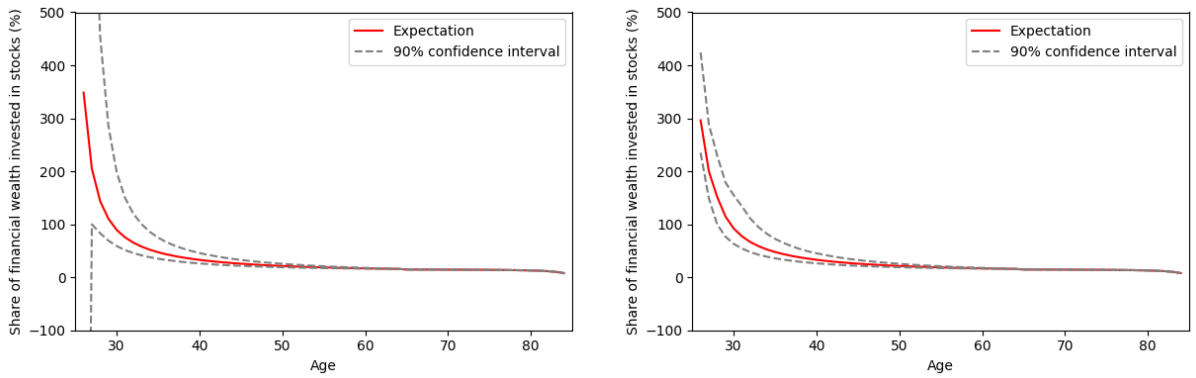
As expected, Table 5.1 shows that all laws result in a welfare loss compared to the optimal situation. With an optimal projected return, relaxing the borrowing constraint from 100% to 150% increases the welfare from 0.911 to 0.915, nearly halving the resulting welfare loss. The projected return equal to the risk-free rate also generates a welfare loss, but this loss is less severe than that associated with the borrowing constraint. For the combination of both laws in the FPA the table reports a welfare loss of roughly 1%, which is higher than all other individual laws. This is as expected, since one law leads to a sub-optimal consumption choice and the other to a sub-optimal stock portfolio choice. Together, these effects compound the overall welfare loss. Since we understand that a higher borrowing constraint would be difficult to justify and implement, the main focus of this research is the determination of the projected return. Table 5.1 shows that with a borrowing constraint of 150%, the optimal calculation of the projected return would decrease the overall welfare loss by more than half.

## 5.2 Life cycle investing with inflation

Now, we consider the model with inflation, using equation 3.13 for the process of the financial wealth. This process adjusts for inflation, so all variables in this section are expressed in real terms unless stated otherwise. Moreover, we use the defined contribution (DC) interpretation in this section, as described in 3.1.4.

### 5.2.1 Stock portfolio

First, we look at the evolution of the stock portfolio in this setting. We compare the optimal situation and the scenario with a borrowing constraint of 150%, as in the FPA. For now, we only look at the result for a correlation $\rho_{S,\pi}$ equal to 0. The graphs are given in Figure 5.8.




(a) Optimal, $\overline{w} = \infty$

(b) $\overline{w} = 1.5$

Figure 5.8: Share of financial wealth in the stock portfolio over time under different borrowing constraints, with inflation and $\rho_{S,\pi} = 0$

Both graphs in Figure 5.8 start at a level far above 100%, but show a more rapidly decreasing share of financial wealth in stocks than in the model without inflation. This does not result from the inclusion of inflation, but is due to the DC interpretation of the model. Since there is no consumption before retirement, the pension pot grows more rapidly during this period. The graphs suggest that the investor prefers to take less risk because of this by investing less of her assets in risky stocks. After retirement, both graphs have the same evolution for all scenarios, which we also saw in the model without inflation. Despite the fact that the investor is allowed to have a negative pension pot with a borrowing constraint of 150%, there are no negative values for the 5% quantile in Figure 5.8b. There are scenarios in which this amount becomes negative, but this is not the case for the best 95% of the scenarios. With a borrowing constraint of 150% and no consumption in the first period, there is a very small chance that the investor ends up with a pension pot below 0 after one year.

As seen in Table 4.1, we consider three different values for $\rho_{S,\pi}$. Most variables only have a small observable difference for each value, except for the stock portfolio. For that reason, we show the evolution of the stock portfolio under different values for the correlation between stock returns and expected inflation, without any restrictions. The graphs are given in Figure 5.9.

The graphs in Figure 5.9 show a similar pattern, with higher expectation levels and more wide confidence intervals for more positive correlation values. At retirement, the expected value of the real amount in the pension pot is highest for a correlation of 0.3, and lowest for a correlation of -0.3. This can be explained by the fact that when stock returns and expected inflation are more positively correlated, the investor is better able to use investments in stocks as a hedge against periods of high inflation. This results in a higher real amount of wealth in the pension pot after some time. When there is a negative correlation, higher stock returns are more often paired with lower inflation rates, which need less hedging. In the periods where there are higher

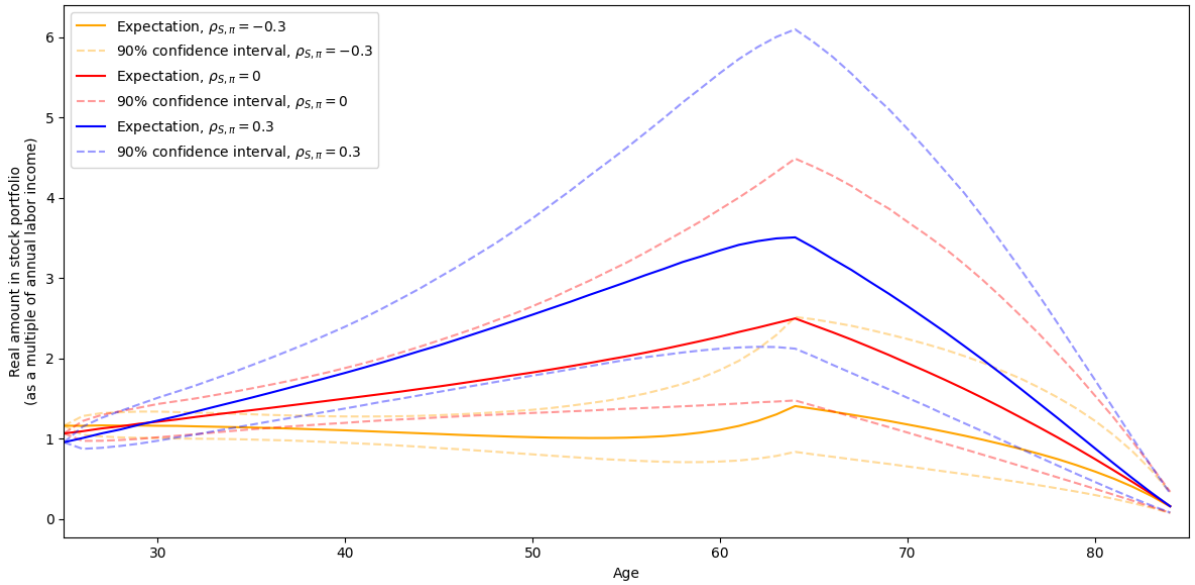



Figure 5.9: Stock portfolio over time under different values for $\rho_{S,\pi}$ and no restrictions

inflation rates, the investor needs to hedge this risk, but is not able to do so because of lower stock returns. This results in a lower expected value for the amount of wealth in the pension pot after some time, almost not exceeding the initial value of the stock portfolio, as seen by the yellow line in Figure 5.9.

### 5.2.2 Consumption

Now, we evaluate how the regulation of the FPA where the risk-free rate is used as the projected return affects the consumption decisions of the investor. We consider the consumption as given, where it is equal to around 0.75 before retirement because of the DC setting, and equal to equation 3.8 with $A = -R_f$ after retirement. We optimize the stock portfolio decision accordingly. Figure 5.10 shows the effects on the consumption. Before retirement, we consider the consumption as the amount of the income that the investor does not put in their pension pot. After retirement, we interpret it as the amount that is distributed to the investor by the pension fund.

Figure 5.10 shows that in both the FPA and the optimal scenario, all given quantiles make a jump at retirement age to different levels of consumption. This is due to the fact that at $t = 0$, the return on the investments during the working life of the investor is uncertain, just as the level of inflation. Higher returns before retirement lead to higher disbursements after retirement. For the worst scenarios, like for the 5% quantiles in Figure 5.10, the consumption after retirement is lower than before retirement. This is caused by underperforming stock investments and/or high levels of inflation.

In the optimal scenario, the level of consumption stays roughly the same throughout retirement. This again is caused by the long-term real interest rate being equal to the rate of time preference, making the investor indifferent in choosing between consuming and investing. With the projected return of the FPA, the consumption starts at a higher and ends at a lower level. This suggests that a projected return equal to the risk-free rate is too high, resulting in excessive

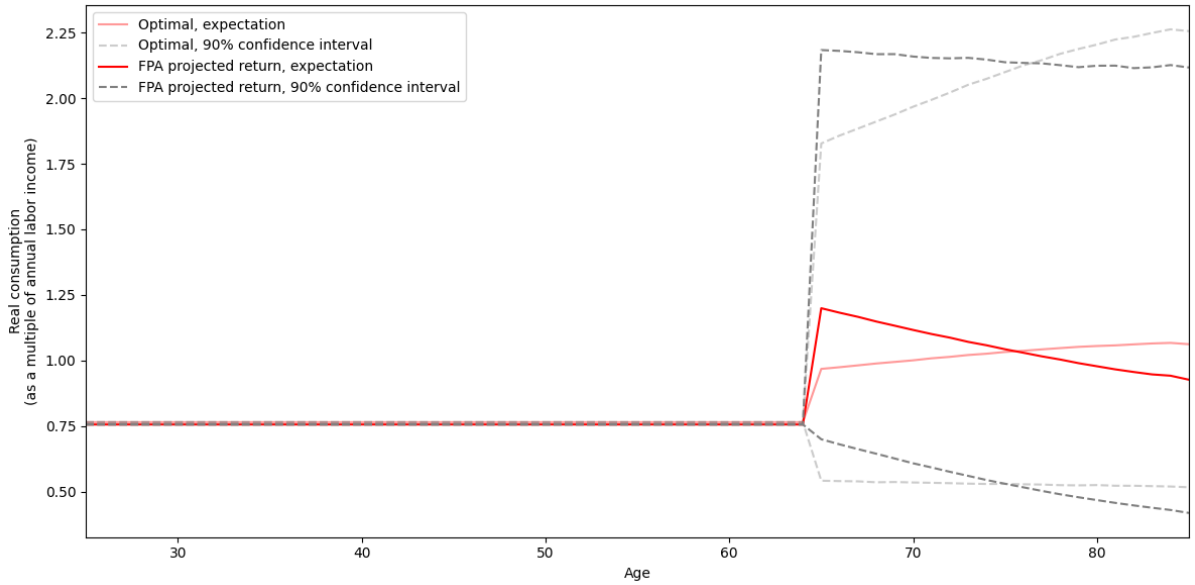



Figure 5.10: Real consumption in the FPA versus the optimal scenario, with $\rho_{S,\pi} = 0$

consumption just after retirement.

In conclusion, due to the sub-optimal projected return of the FPA, the investor receives a decreasing amount in real terms during her retirement, creating less purchasing power. By relaxing this regulation, pension funds would be better able to smooth out their disbursements for their participants.

### 5.2.3 Financial wealth

Lastly, we evaluate the evolution of the real amount of wealth in the pension pot of the investor. This variable is affected by both decision variables and all FPA regulations. In Figure 5.11 below we first analyze the process of the pension pot in the optimal situation, without any regulations and with no correlation between stock returns and expected inflation.

The graph shows, just like in the model without inflation, a pattern that gradually increases before retirement, and decreases towards zero after retirement. In the first years the pension pot does become negative for some simulation paths, but this is only for a short time and a small amount. We see that the confidence interval in Figure 5.11 is very narrow in the early years compared to the model without inflation. This is a result of DC, since the investor has no flexibility in changing their consumption decision and hence they take less leverage at a younger age. This also causes the confidence interval to be more wide at retirement, but this also results from the inflation. With the two additional random variables after accounting for inflation, there is more uncertainty in this model. The decrease towards zero after retirement is less convex than for the model without inflation, best observable when looking at the 5% quantile. This results from the fact that the long-term real interest rate, which is calculated by subtracting the long-term inflation rate from the interest rate, is equal to the rate of time preference. This makes the average investor indifferent between saving and consuming, because the marginal utility of present consumption equals the expected discounted marginal utility of future consumption.

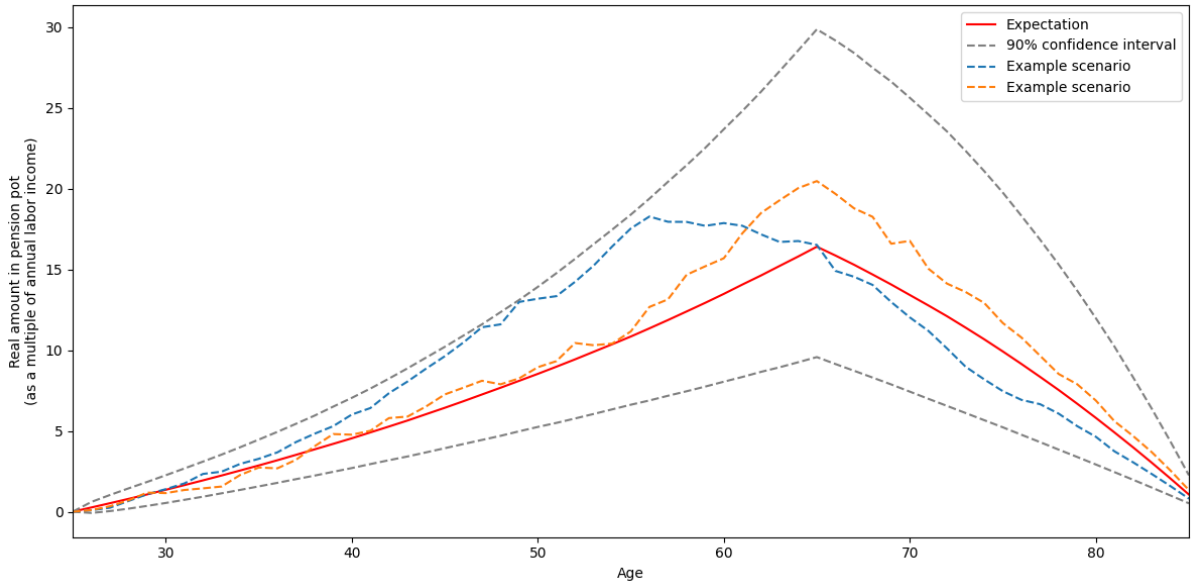



Figure 5.11: Real amount of wealth in the pension pot with $\rho_{S,\pi} = 0$ and no restrictions

Next, we analyze how both FPA regulations affect the evolution of the real amount of wealth in the pension pot. Figure 5.12 shows the graphs for the optimal scenario and the FPA setting.

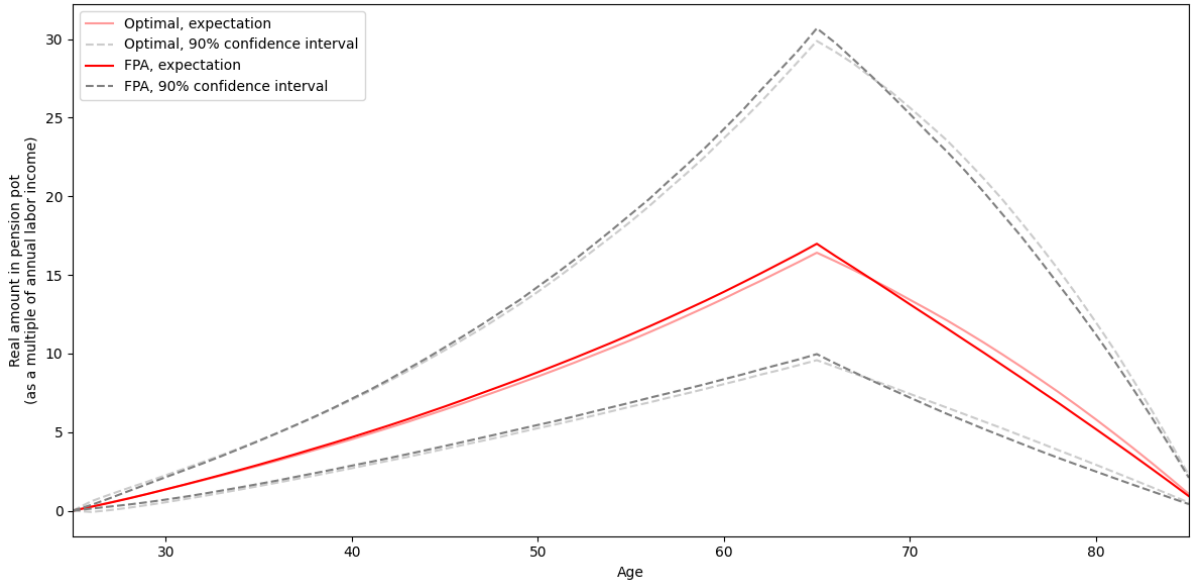



Figure 5.12: Real amount of wealth in the pension pot with $\rho_{S,\pi} = 0$ in the FPA versus the optimal situation

The graphs in Figure 5.12 show almost no difference. The pension contribution before retirement in the FPA is slightly higher (24.4%) than in the optimal situation (23.5%), resulting in a higher amount of wealth in the pension pot at retirement in the FPA. There is a difference between the confidence intervals of the two graphs in the first years, but as the figure shows, this difference is very small. We already saw this earlier in Figure 5.8, where we concluded that the borrowing constraint of 150% has less effect on the stock portfolio decision of the investor

in a DC setting. The only difference before retirement in Figure 5.12 is that the borrowing constraint causes the investor to take less risk, making the confidence interval more narrow. After retirement, we see that under the FPA the financial wealth decreases faster towards zero. This is due to the excessive consumption just after retirement, resulting from a projected return equal to the risk-free rate.

In conclusion, the borrowing constraint has almost no effect on the purchasing power of the investor at retirement. This holds for all three values for the correlation. Mostly due to the sub-optimal projected return, the purchasing power of the investor decreases during retirement.

### 5.2.4 Welfare loss

In this section we discuss the welfare loss under different sub-optimal laws of the FPA. We again consider the optimal scenario, one with a borrowing constraint of 100% and one with 150%, the scenario with the projected return equal to the risk-free rate, and the combination of the last two. Table 5.2 shows the results. The welfare is reported in the level of consumption that is equivalent to the average utility of the investor.

Table 5.2: Welfare levels expressed as consumption equivalents under different regulations of the FPA, accounting for inflation and different values for the correlation between stock returns and expected inflation

| CE | Optimal | 100% BC | | 150% BC | | RTS PR | | RTS PR & 150% BC | |
|---|---|---|---|---|---|---|---|---|---|
| $\rho_{S,\pi} = -0.3$ | 0.751 | 0.750 | (-0.06%) | 0.750 | (-0.04%) | 0.739 | (-1.59%) | 0.738 | (-1.63%) |
| $\rho_{S,\pi} = 0$ | 0.761 | 0.760 | (-0.06%) | 0.761 | (-0.03%) | 0.750 | (-1.40%) | 0.750 | (-1.44%) |
| $\rho_{S,\pi} = 0.3$ | 0.776 | 0.775 | (-0.06%) | 0.775 | (-0.03%) | 0.767 | (-1.10%) | 0.767 | (-1.13%) |

**Note:** CE = consumption equivalent; BC = borrowing constraint; RTS PR = projected return equals RTS of the DNB; $\rho_{S,\pi}$ = correlation between stock returns and expected inflation.

The levels of certainty equivalent consumption in Table 5.2 are lower than for the model without inflation. This is as expected, since adding inflation when all other parameters remain equal, has a negative effect on the level of wealth and hence consumption of the investor.

Table 5.2 shows a welfare loss for all regulations, emphasizing the sub-optimality of the laws. A more negative correlation between stock returns and expected inflation results in a lower welfare level, caused by the lower investment returns due to the impaired ability to hedge against inflation, as we have seen previously. Table 5.2 also shows that as the correlation becomes more positive, the welfare loss decreases, implying that the FPA regulations become less restrictive, allowing investors to act more in line with their optimal, unconstrained strategy. For the borrowing constraint, this is caused by the fact that when the investor has less wealth due to poor returns, caused by a more negative correlation, the marginal utility of consumption is higher. For this the investor needs to borrow more, but this is restricted by the borrowing constraint. Moreover, a more positive correlation results in a faster accumulation of wealth, making her less likely to hit the constraint. For the projected return, we know that the risk-free rate is higher than the true projected return, because it does not take future inflation into account. When the correlation is more positive, the expected future returns are higher, shifting the true projected return more towards the risk-free rate. This makes the FPA projected return less sub-optimal, resulting in a lower welfare loss for a more positive correlation.

The relaxation of the borrowing constraint to 150% reduces the welfare loss between 40% and 50%. However, the welfare loss due to the borrowing constraint is a lot smaller compared to the model without inflation. This emphasizes the fact that the borrowing constraint has less to almost no effect in the DC setting. In contrast to the results for the model without inflation, the sub-optimal projected return now accounts for a larger welfare loss than the borrowing constraint. For the FPA projected return the welfare loss has increased, because the inflation is an extra component in the true projected return that is not accounted for in the FPA.

Table 5.2 reports that for the combination of the two FPA laws, the welfare loss is between 1% and 2%, depending on the correlation. This is a substantial percentage in the context of Dutch pensions. For comparison, Van Ewijk et al. (2017) consider a welfare loss of just 0.5% already to be substantial. If pension funds are able to account for inflation, time preference and returns on risky investments in the determination of the projected return, the welfare loss would be reduced from approximately 1.40% to 0.03%, keeping the borrowing constraint at 150%. For a more negative correlation between stock returns and expected inflation, this potential gain in welfare increases. This can be partly realized by a fixed mark-up or markdown on the projected return by the pension fund in the current legislation. However, large pension funds like ABP choose not to use this (ABP, Raad voor het Overheidspersoneelsbeleid, 2024), and to fully circumvent this welfare loss, the FPA legislation should be revised to permit a time-varying markup or markdown on the projected return based on expected inflation.

# Chapter 6

# Conclusion

In this paper, we numerically solved a life cycle investing problem in the context of the Dutch Future Pensions Act (FPA). We evaluated how the FPA regulations affect the decisions, overall welfare and purchasing power of the investor in a defined contributions setting. The regulations we consider are the alleviation of the borrowing constraint from 100% to 150% and taking the projected return equal to the risk-free rate. First, we considered the classic Merton (1971) model without inflation. Our numerical optimization technique provided very similar results to the analytical solution. Second, we extended the model by incorporating expected and realized inflation in a defined contributions (DC) setting. For the correlation between stock returns and expected inflation we used multiple values, to see if our results hold for the true value of the parameter. We aimed to see how much welfare is gained when a pension fund would be able to take expected inflation rates into account when disbursing to their participants after retirement.

In Monte Carlo simulations of our optimal model, we have found that with a borrowing constraint of 150% a welfare loss of a substantial 1.4% can be circumvented when using an optimal projected return. This projected return of the FPA should take time preference, returns on risky investments and most of all expected inflation into account. A fixed markup or markdown on the projected return is able to reduce the welfare partly in the current legislation, but it can be fully avoided by allowing to make this time-varying based on expected inflation in the FPA. We also found that in a DC setting with inflation and optimal projected return, the incentive of the investor to fully utilize the borrowing constraint is low. This results in a welfare loss of only 0.03% compared to a situation without a borrowing constraint. Nevertheless, the welfare loss resulting from the borrowing constraint is reduced by almost half after relaxing it from 100% to 150%. Our findings show that the impact of the borrowing constraint on the pension participant's purchasing power is relatively small, especially when contrasted with the much larger effect of the FPA's suboptimal projected return regulation.

## Limitations

This analysis is based on several simplifying assumptions, which may affect the generalizability of the results.

First, the parameters used are broadly aligned with economic intuition, but were primarily chosen for tractability. This limits their representativeness of real-world scenarios, because they

are not calibrated to empirical data of the Dutch market. This also holds for the stochastic processes, which are based on other life cycle investing literature. For future research, other utility functions or other distributions for the random variables in the model could be considered to capture the complexity of economic behavior.

Second, our numerical optimization techniques rely on several approximations. For example, we use linear interpolation for non-linear relations. However, this had little impact as seen by our smooth graphs, and a better way of interpolating would substantially increase computational time.

Third, following other life cycle investing literature, we could account for more randomness in our model. For example, we could consider the life expectancy, annual labor income or the interest rate as a random process. While these are good suggestions for future research, we have chosen to consider these variables as given, since this reduces computational time.

Lastly, while the implementation is not trivial and most pension funds choose to keep it simple (see for example ABP, Raad voor het Overheidspersoneelsbeleid (2024)), there are some additional instruments in the FPA that are able to protect against periods of high inflation. For example, a fixed mark-up or markdown can be used for the projected return by a pension fund, or the solidarity reserve can be used in periods of high inflation (but not expected inflation). We have briefly investigated optimizing the mark-up or markdown using a brute force method, and found that the welfare loss can be reduced to roughly 0.3%. We recommend further research on this topic.

In summary, while these limitations impose some constraints on the generalizability of our findings, they also point to important directions for future research.

## Acknowledgments

I thank my internship supervisor Roel Mehlkopf and thesis supervisor Rutger-Jan Lange for their insights, comments and great supervision. I also thank people at Cardano Risk Management in general for providing an internship where I was able to grow on a professional level, as well as the freedom to work individually on my thesis.

For my research I have used AI (OpenAI, 2023) to make my code Numba (Lam et al., 2015) compatible and provide suggestions on writing.

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# Appendix A

# Proofs

## A.1 Proof equation 3.19

Gauss-Hermite quadrature is built for integrals of the form

$$\int_{-\infty}^{+\infty} f(x)e^{-x^2}dx \approx \sum_{i=1}^{n} w_i f(x_i). \tag{A.1}$$

Using the definition of $G$ and $\varepsilon$ from equation 3.18, we get

$$\begin{aligned}
\mathbb{E}[f(\mathbf{Z})] &= \mathbb{E}[f(G\varepsilon)] \\
&= \int_{\mathbb{R}^3} f(G\varepsilon)\phi(\varepsilon_S)\phi(\varepsilon_\Pi)\phi(\varepsilon_\pi)d\varepsilon \\
&= \int_{\mathbb{R}^3} f(G\varepsilon)\frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}\varepsilon_S^2}d\varepsilon_S\frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}\varepsilon_\Pi^2}d\varepsilon_\Pi\frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}\varepsilon_\pi^2}d\varepsilon_\pi,
\end{aligned} \tag{A.2}$$

where $\phi(\cdot)$ is the standard normal PDF. The integral needs a change of variables to match the form of A.1. Using $x = \frac{\varepsilon}{\sqrt{2}} \iff \varepsilon = \sqrt{2}x$, we get

$$\begin{aligned}
\mathbb{E}[f(\mathbf{Z})] &= \int_{\mathbb{R}^3} f\left(G\sqrt{2}\mathbf{x}\right)\frac{1}{\sqrt{\pi}}e^{-x_S^2}dx_S\frac{1}{\sqrt{\pi}}e^{-x_\Pi^2}dx_\Pi\frac{1}{\sqrt{\pi}}e^{-x_\pi^2}dx_\pi \\
&\approx \frac{1}{\pi^{3/2}}\sum_{i=1}^{n}\sum_{j=1}^{n}\sum_{k=1}^{n} w_i w_j w_k f\left(\sqrt{2}G\mathbf{x}\right).
\end{aligned} \tag{A.3}$$

# Appendix B

# Extra figures

This appended chapter shows some extra images that are generated from our models. First, we show the non-linear relation between future financial wealth, future expected inflation and future consumption equivalent for a specific model and time period. Then, we show three graphs for each model: (real) financial wealth, (real) consumption and share of financial wealth invested in the stock portfolio. First we show the model without inflation, and afterwards the model with inflation with the three different values for the correlation between stock returns and expected inflation. We show each variable for different FPA regulations.

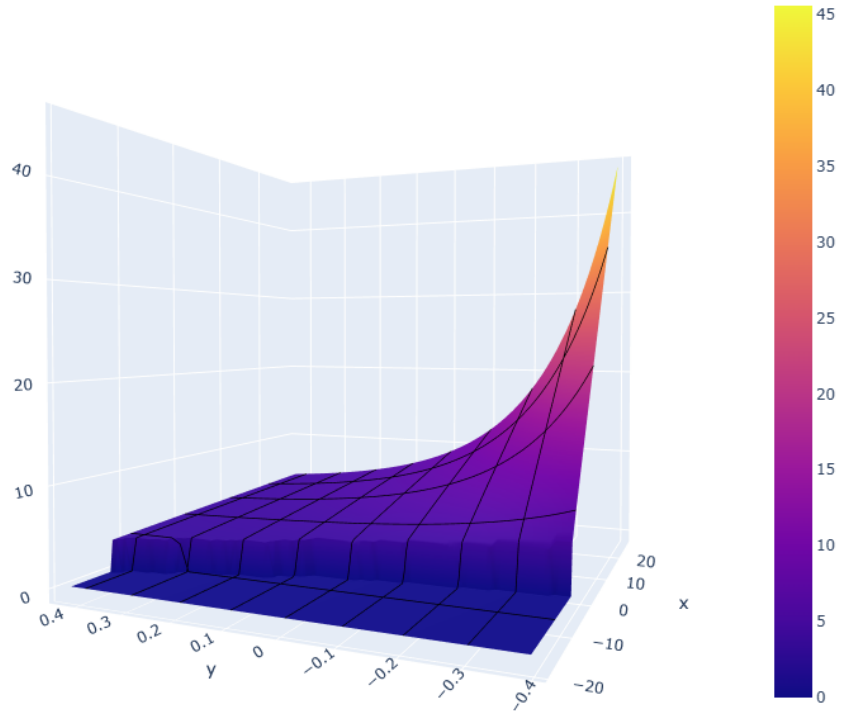
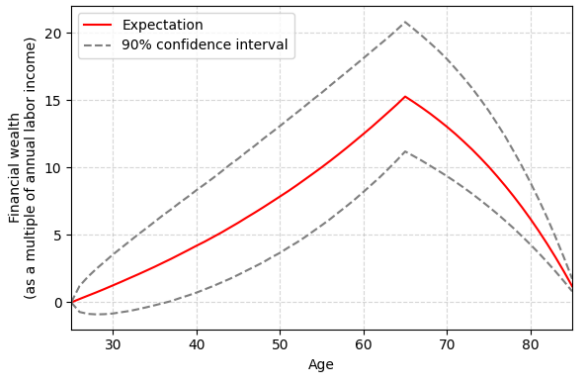
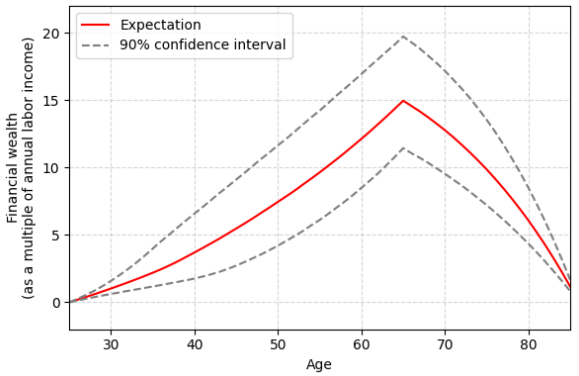


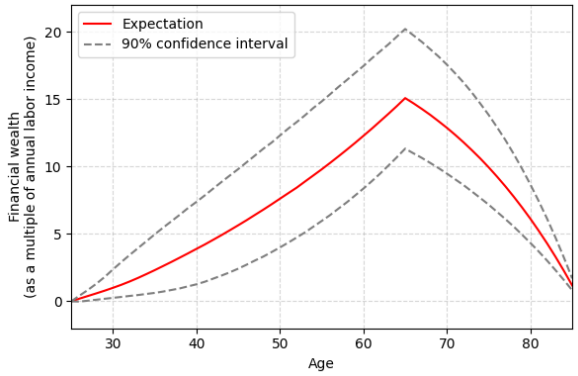

Figure B.1: Relation between future financial wealth (x), future expected inflation (y, as a fraction of 1) and future consumption equivalent (z) for model with inflation, no regulations, $\rho_{S,\pi} = 0$ and $t = 0$



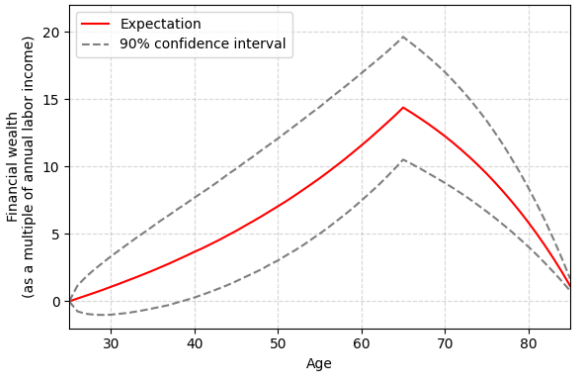

(a) Optimal



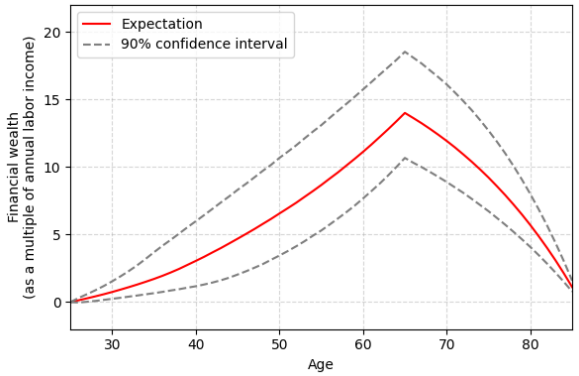

(b) $\overline{w} = 1$




(c) $\overline{w} = 1.5$

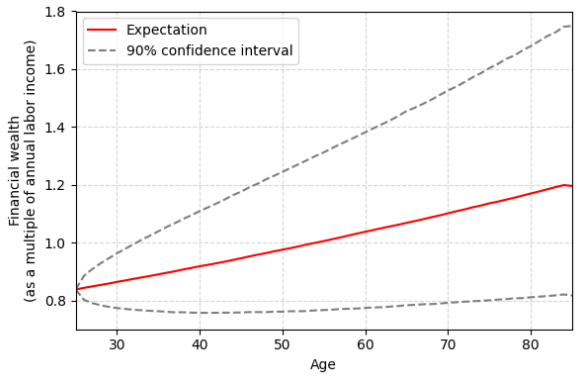



(d) $A = -R_f$

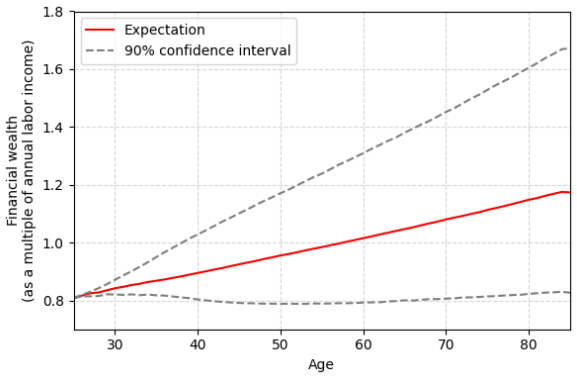



(e) $\overline{w} = 1.5$ and $A = -R_f$

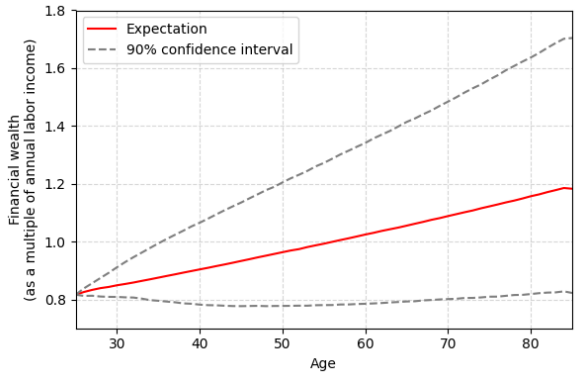
Figure B.2: Financial wealth for model without inflation under different regulations



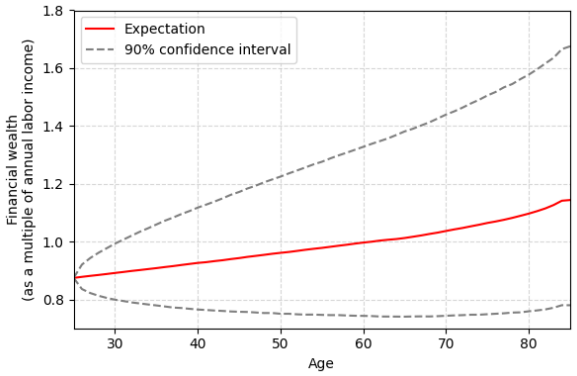

(a) Optimal



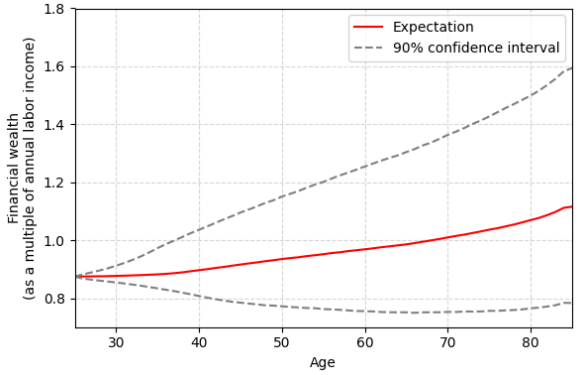

(b) $\overline{w} = 1$




(c) $\overline{w} = 1.5$

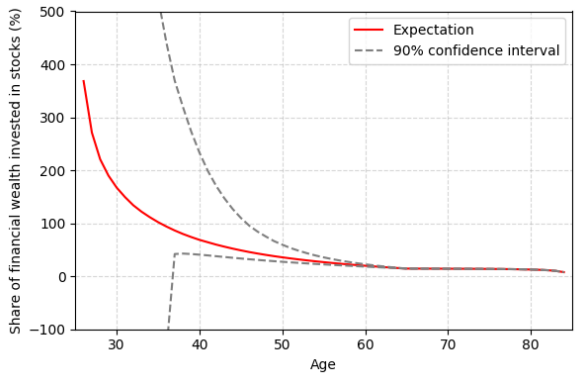



(d) $A = -R_f$

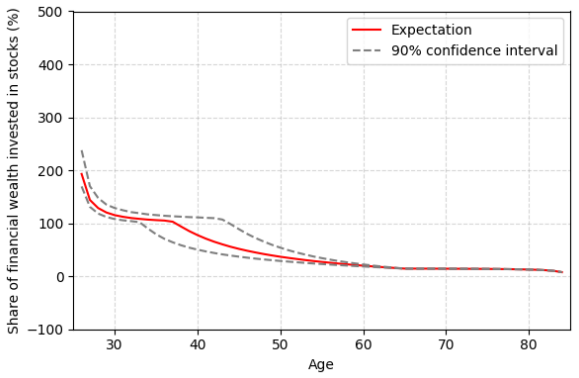



(e) $\overline{w} = 1.5$ and $A = -R_f$

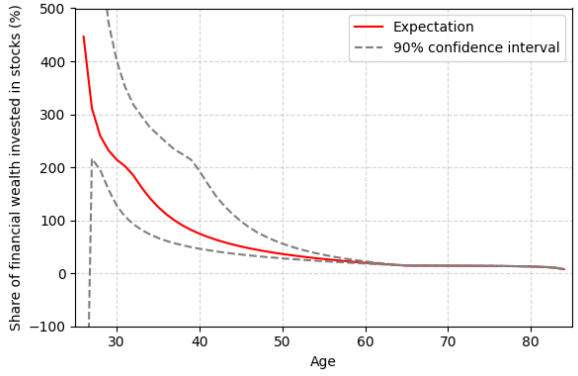
Figure B.3: Consumption for model without inflation under different regulations



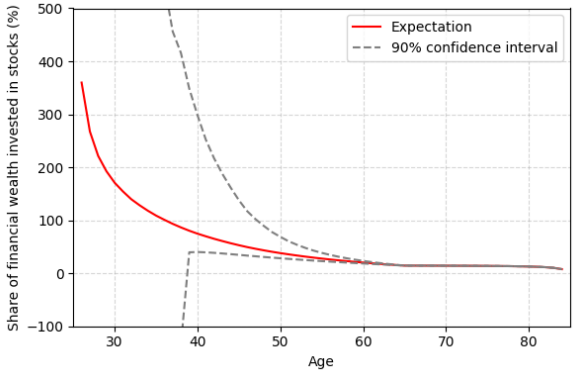

(a) Optimal



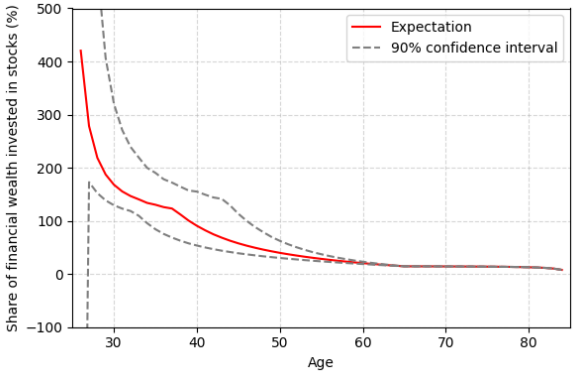

(b) $\overline{w} = 1$




(c) $\overline{w} = 1.5$

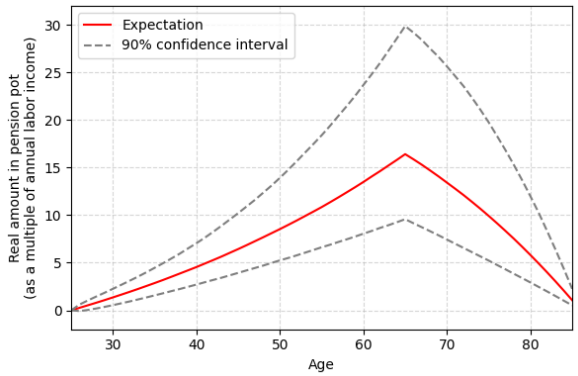



(d) $A = -R_f$

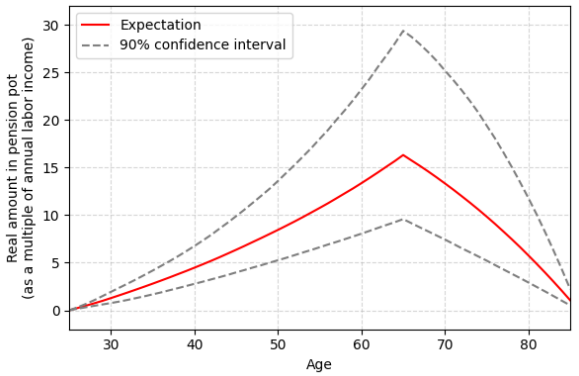



(e) $\overline{w} = 1.5$ and $A = -R_f$

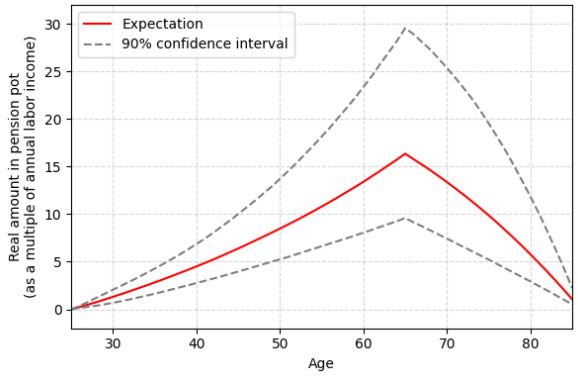
Figure B.4: Share of financial wealth invested in stocks for model without inflation under different regulations



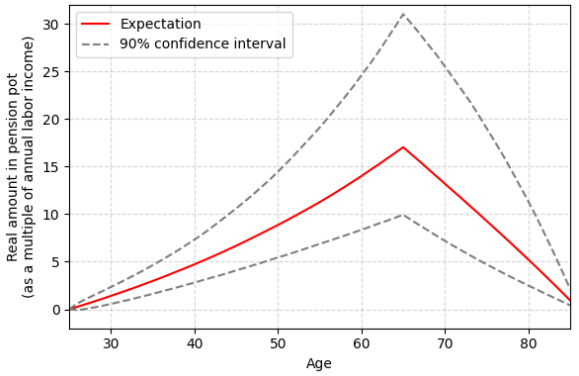

(a) Optimal



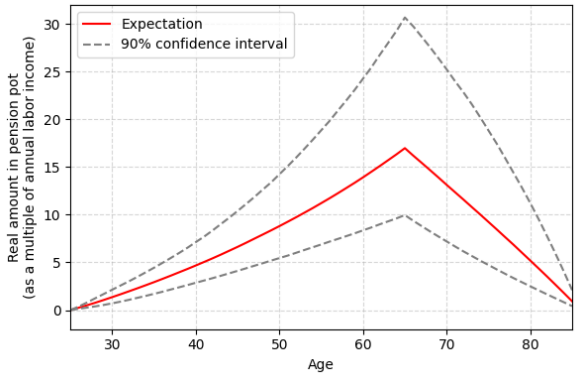

(b) $\overline{w} = 1$




(c) $\overline{w} = 1.5$

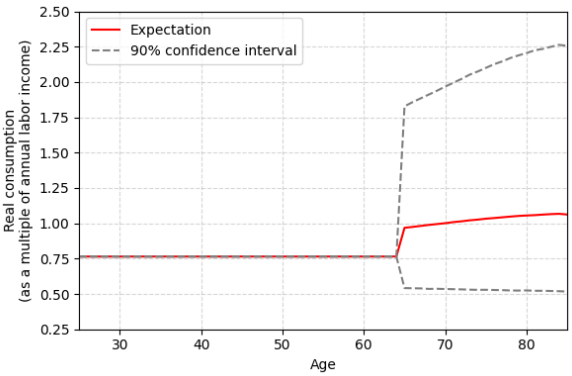



(d) $A = -R_f$

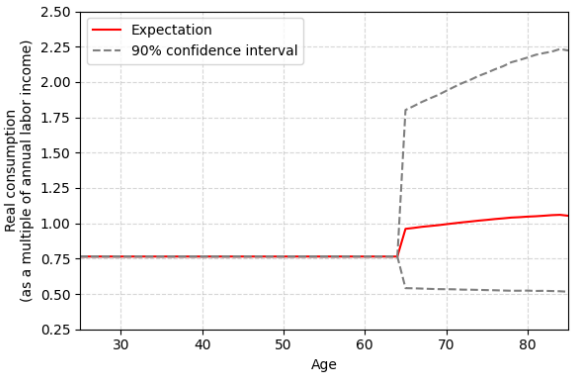



(e) $\overline{w} = 1.5$ and $A = -R_f$

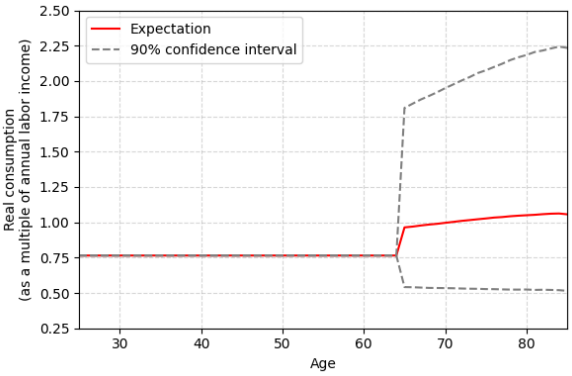
Figure B.5: Financial wealth for model with inflation under different regulations, $\rho_{S,\pi} = 0$



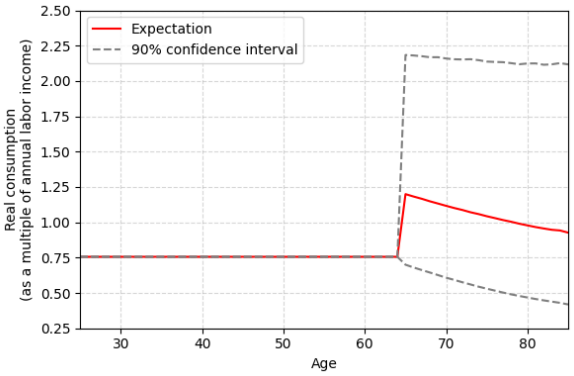

(a) Optimal



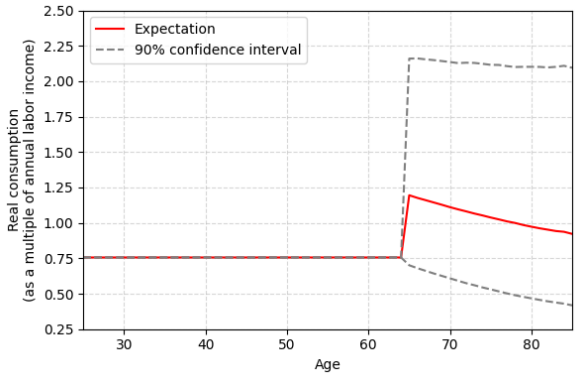

(b) $\overline{w} = 1$




(c) $\overline{w} = 1.5$

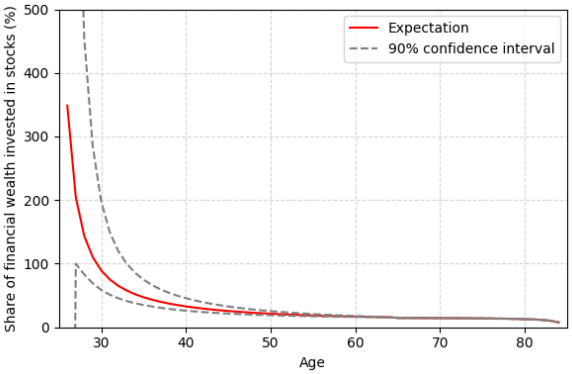



(d) $A = -R_f$

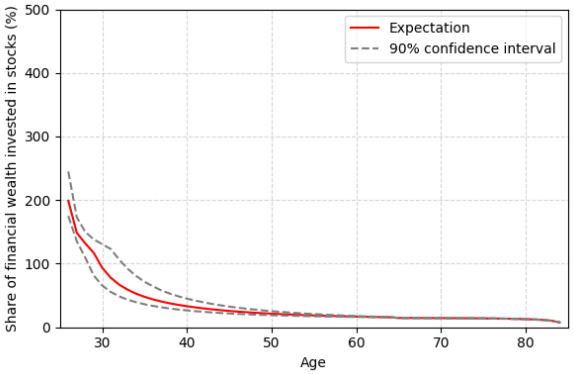



(e) $\overline{w} = 1.5$ and $A = -R_f$

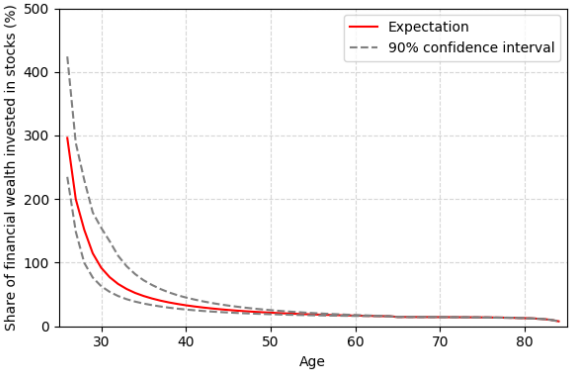
Figure B.6: Consumption for model with inflation under different regulations, $\rho_{S,\pi} = 0$



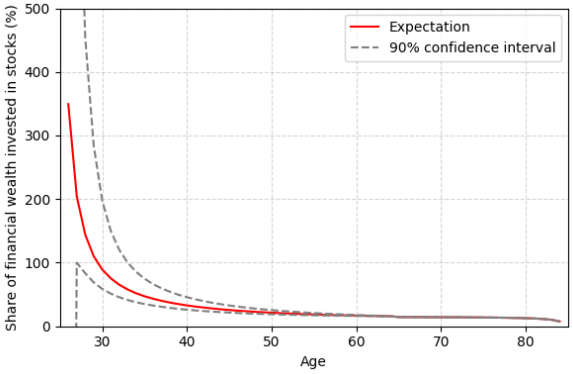

(a) Optimal



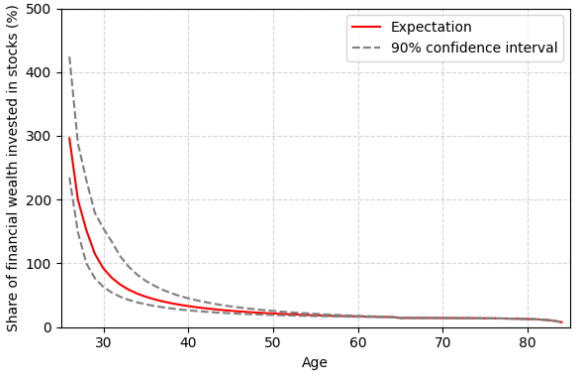

(b) $\overline{w} = 1$




(c) $\overline{w} = 1.5$

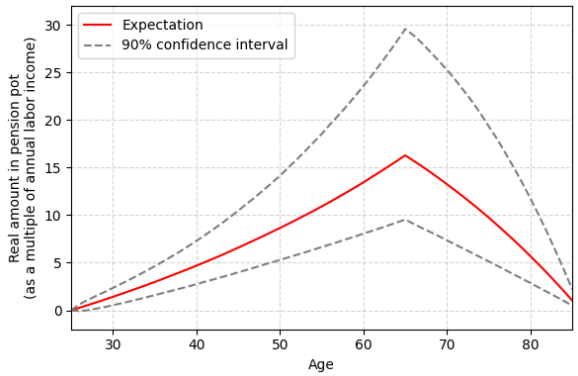



(d) $A = -R_f$

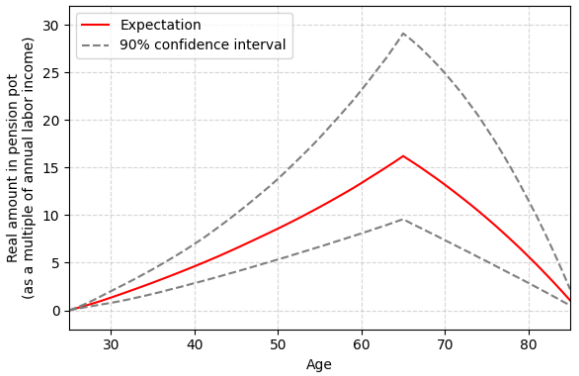



(e) $\overline{w} = 1.5$ and $A = -R_f$

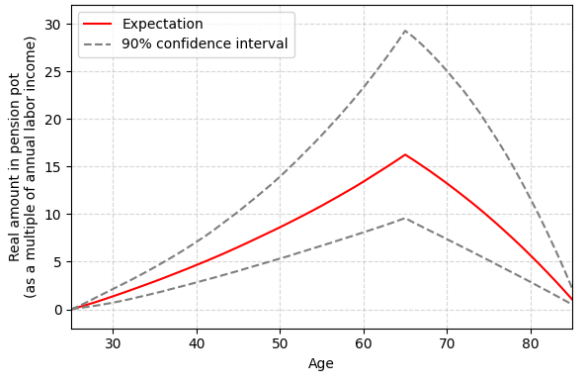
Figure B.7: Share of financial wealth invested in stocks for model with inflation under different regulations, $\rho_{S,\pi} = 0$



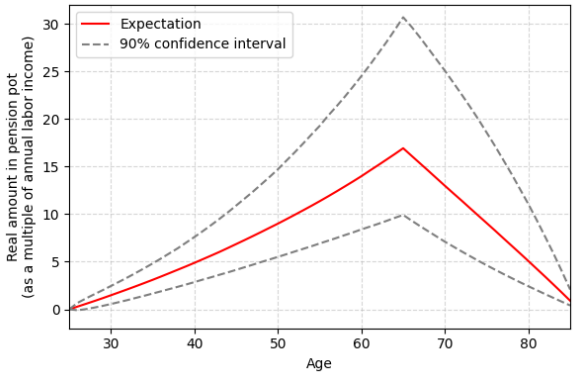

(a) Optimal



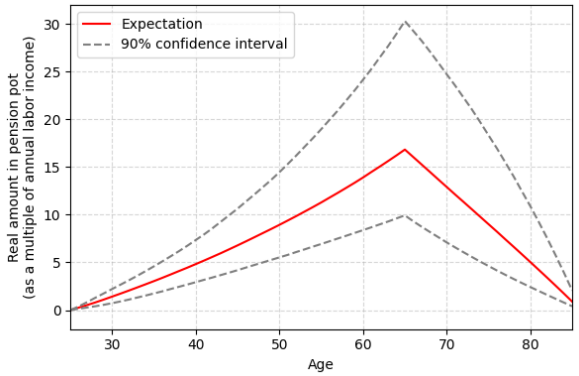

(b) $\overline{w} = 1$




(c) $\overline{w} = 1.5$

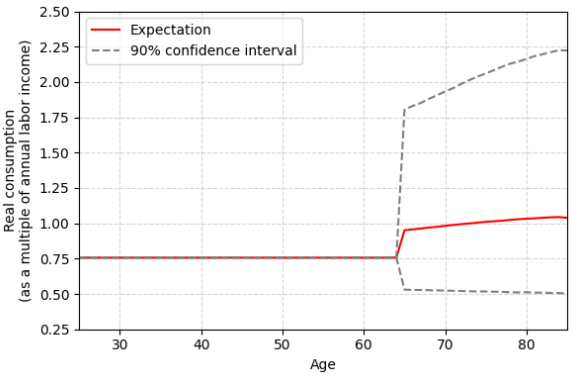



(d) $A = -R_f$

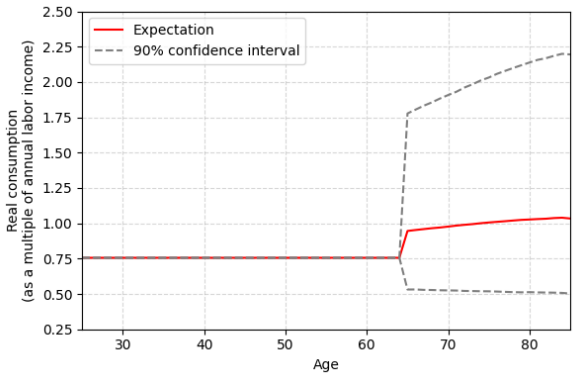



(e) $\overline{w} = 1.5$ and $A = -R_f$

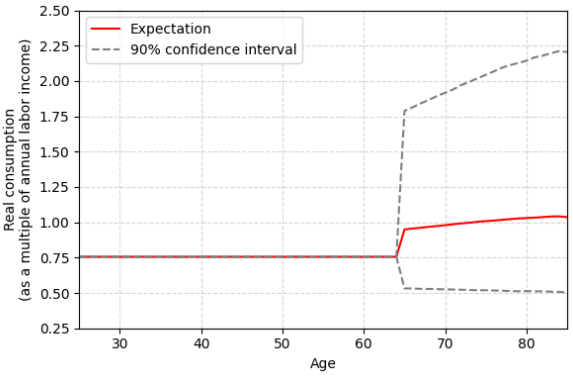
Figure B.8: Financial wealth for model with inflation under different regulations, $\rho_{S,\pi} = -0.3$



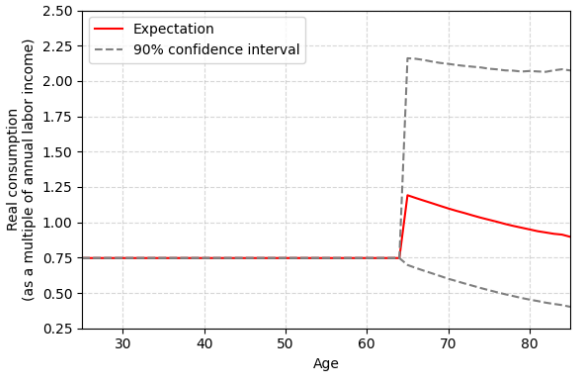

(a) Optimal



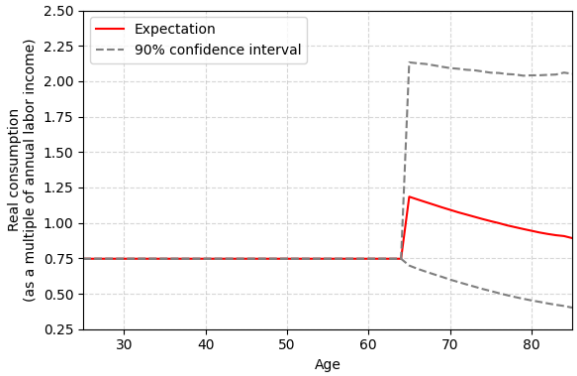

(b) $\overline{w} = 1$




(c) $\overline{w} = 1.5$

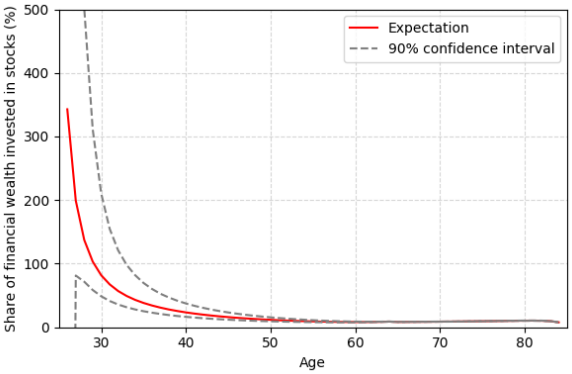



(d) $A = -R_f$

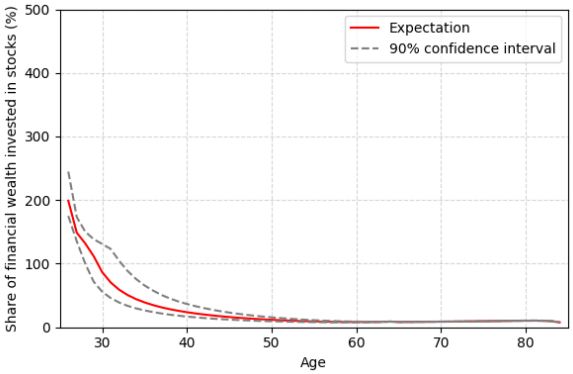



(e) $\overline{w} = 1.5$ and $A = -R_f$

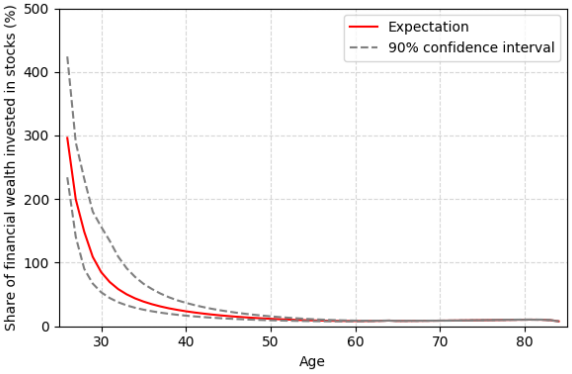
Figure B.9: Consumption for model with inflation under different regulations, $\rho_{S,\pi} = -0.3$



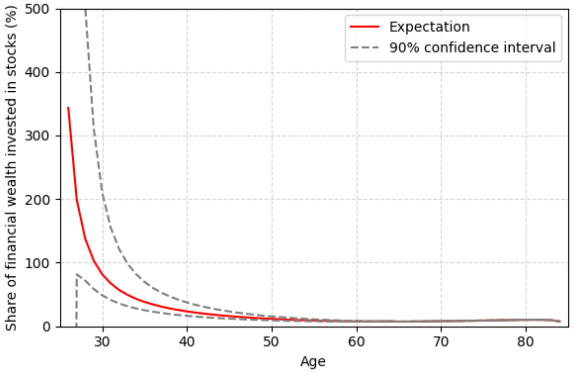

(a) Optimal



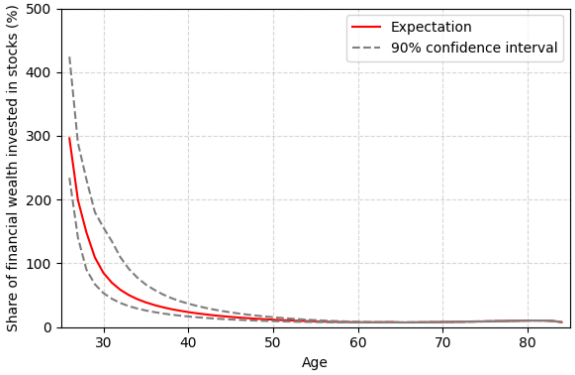

(b) $\overline{w} = 1$




(c) $\overline{w} = 1.5$

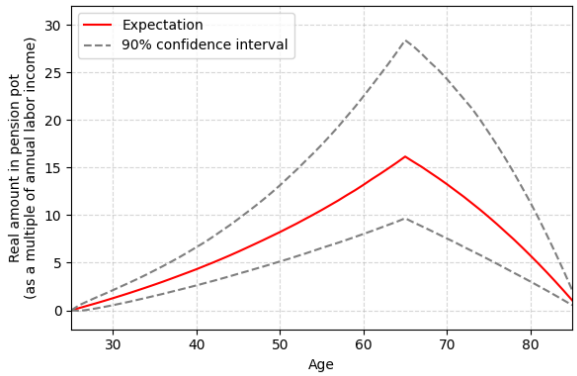



(d) $A = -R_f$

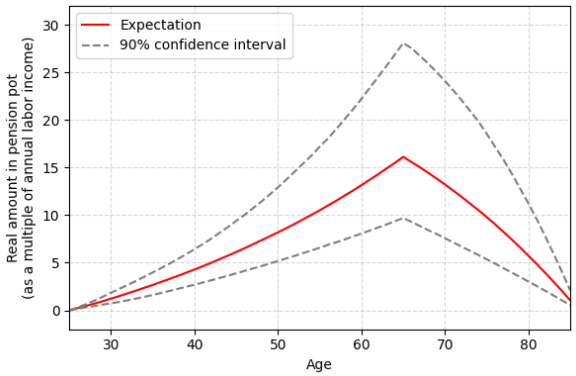



(e) $\overline{w} = 1.5$ and $A = -R_f$

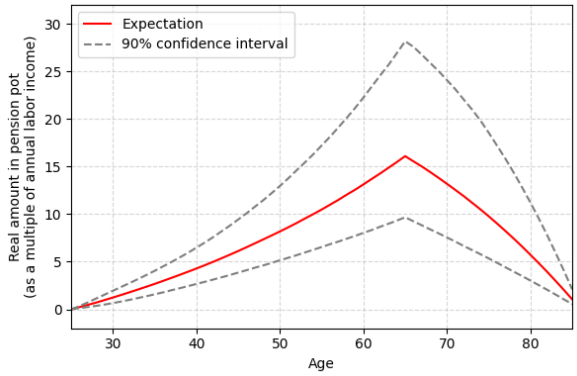
Figure B.10: Share of financial wealth invested in stocks for model with inflation under different regulations, $\rho_{S,\pi} = -0.3$



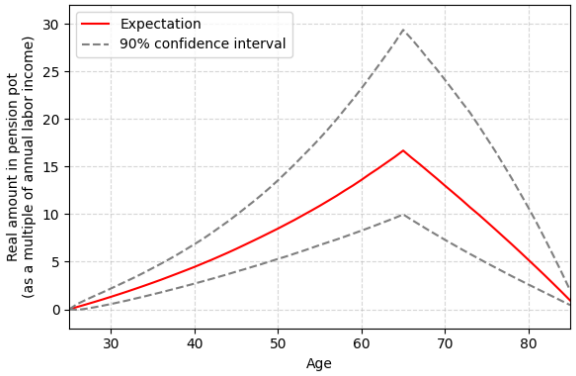

(a) Optimal



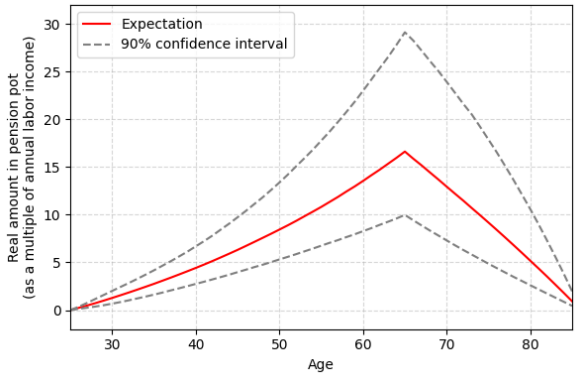

(b) $\overline{w} = 1$




(c) $\overline{w} = 1.5$

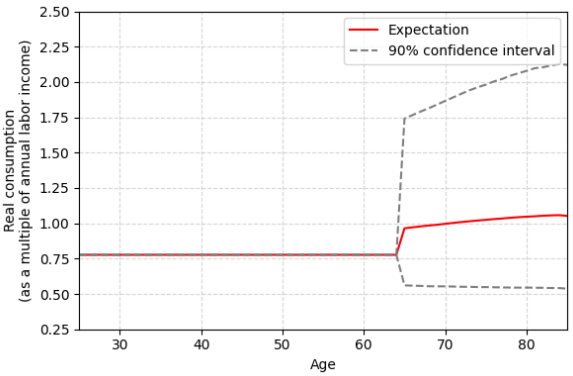



(d) $A = -R_f$

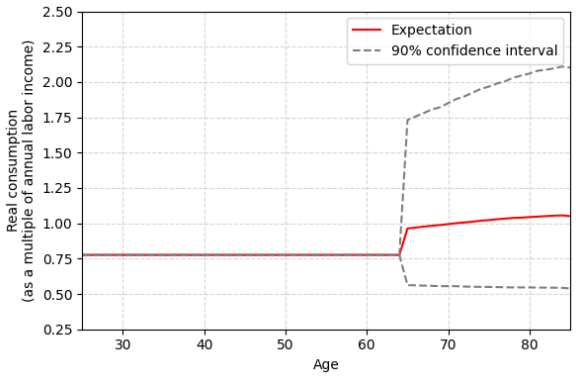



(e) $\overline{w} = 1.5$ and $A = -R_f$

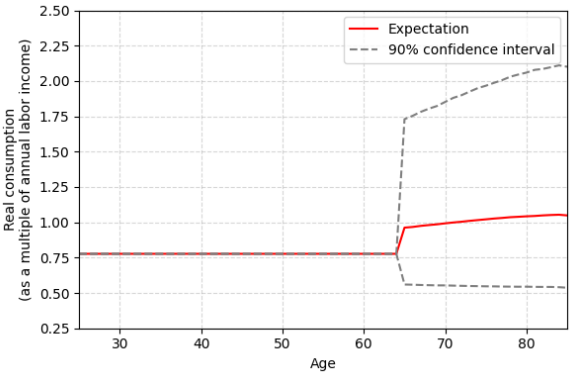
Figure B.11: Financial wealth for model with inflation under different regulations, $\rho_{S,\pi} = 0.3$



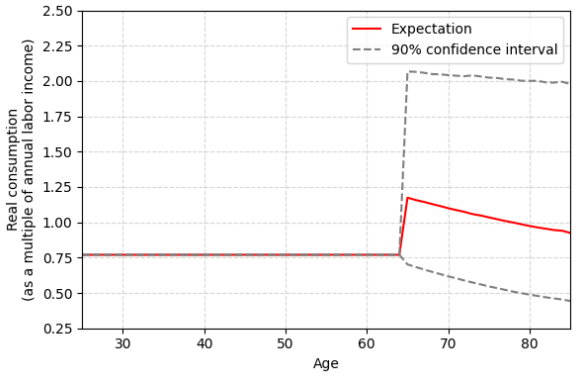

(a) Optimal



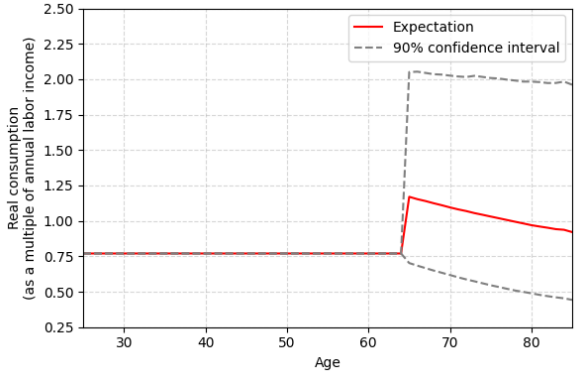

(b) $\overline{w} = 1$




(c) $\overline{w} = 1.5$

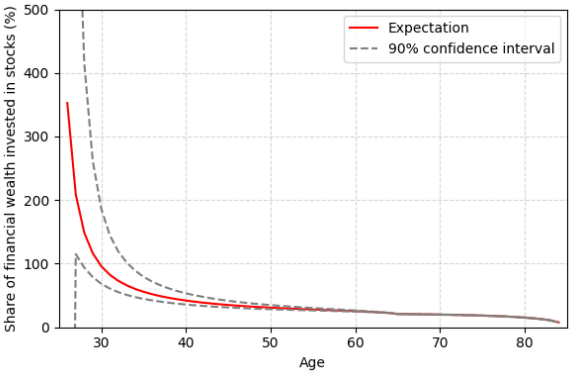



(d) $A = -R_f$

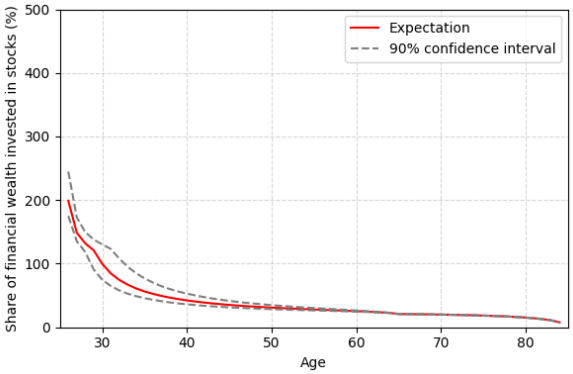



(e) $\overline{w} = 1.5$ and $A = -R_f$

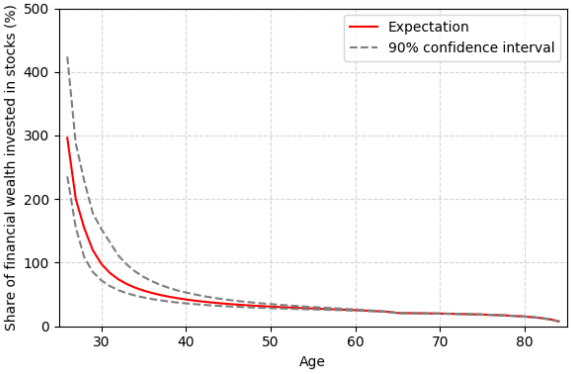
Figure B.12: Consumption for model with inflation under different regulations, $\rho_{S,\pi} = 0.3$



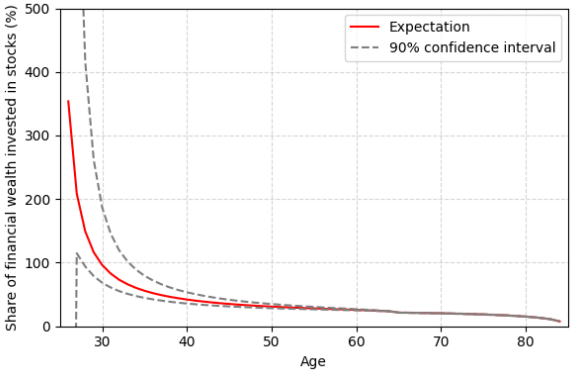

(a) Optimal



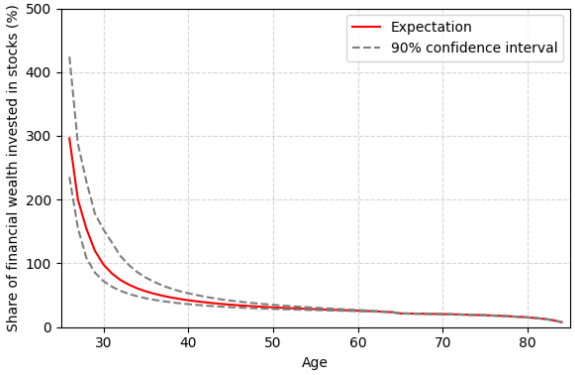

(b) $\overline{w} = 1$




(c) $\overline{w} = 1.5$




(d) $A = -R_f$




(e) $\overline{w} = 1.5$ and $A = -R_f$

Figure B.13: Share of financial wealth invested in stocks for model with inflation under different regulations, $\rho_{S,\pi} = 0.3$